

Dark matter and galaxy cross-correlations with the Cherenkov Telescope Array Observatory

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(Received 13 June 2025; accepted 29 October 2025; published 4 December 2025)

The Cherenkov Telescope Array Observatory (CTAO) will be a ground-based Cherenkov telescope performing wide-sky surveys, ideal for anisotropy studies such as cross-correlations with tracers of the cosmic large-scale structure. Cross-correlations can shed light on high-energy γ -ray sources and potentially reveal exotic signals from particle dark matter. In this work, we investigate CTAO sensitivity to cross-correlation signals between γ -ray emission and galaxy distributions. We find that by using dense, low-redshift catalogs like 2 MASS, and for integration times around 50 hours, this technique achieves sensitivities to both annihilating and decaying dark matter signals that are competitive with those from dwarf galaxy and cluster analyses.

DOI: [10.1103/PhysRevD.112.123010](https://doi.org/10.1103/PhysRevD.112.123010)

I. INTRODUCTION

High-energy cosmic messengers offer a glimpse into some of the most powerful phenomena in the Universe. Among these messengers, γ -rays stand out as a unique tool for exploring the physical processes driving astrophysical sources such as active galactic nuclei (AGNs), supernovae, star-forming galaxies, and pulsars. At the same time, γ -rays are among the most promising observational signatures expected from dark matter (DM), should it consist of a new particle capable of self-annihilation or decay.

The Cherenkov Telescope Array Observatory (CTAO) [1,2], designed to detect cosmic γ -rays with energies above 20 GeV, will offer a unique opportunity to probe both the

high-energy emission of astrophysical sources and potential signals from DM particles with masses in the TeV range. CTAO will improve upon the performance of current imaging atmospheric Cherenkov telescopes, thanks to its better angular resolution, larger field of view, and broader energy coverage. It will also be the first ground-based Cherenkov telescope to carry out an intensive program of dedicated sky surveys [1,2], covering large regions (i.e. more than a quarter) of the sky. These capabilities make CTAO particularly well suited to studying the cosmological γ -ray emission potentially produced by DM annihilation or decay. However, this emission is irreducibly intermingled with the nonthermal emissions from astrophysical sources. The diffuse unresolved γ -ray background (UGRB) consists of the cumulative emission from γ -ray astrophysical sources which are too faint to be individually resolved, with a possible additional contribution from DM [3].

A potentially powerful method to separate the astrophysical emission from a DM contribution is the analysis of anisotropies in the γ -ray sky. Various techniques are possible for this purpose. The present work focuses in particular on the autocorrelation of the gamma-ray signal

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and cross-correlations between the γ -ray emission and a gravitational tracer of the DM distribution in the Universe. Autocorrelation analysis are especially well-suited for studying the properties of rare and bright astrophysical populations, like blazars, as opposed to faint and numerous populations, such as star-forming galaxies [4–8]. The cross-correlation technique, instead, offers the advantage of incorporating direct gravitational tracers of the matter distribution in the Universe, and can leverage differences in angular scale, energy spectrum, and redshift evolution between the cross-correlated fields. [9–11] This approach provides a multidimensional handle to disentangle signals with distinct features in their spatial distribution, energy spectra, and redshift dependence. Extragalactic astrophysical sources, for instance, typically appear as pointlike emitters at γ -ray energies, whereas DM is expected to produce a diffuse signal that traces the large-scale structure of the Universe. Different classes of astrophysical sources exhibit diverse spectral energy distributions, which are generally distinct from those predicted for DM-induced emissions. Moreover, their redshift distributions differ; DM γ -ray emission is expected to peak at low redshift, unlike unresolved astrophysical populations [9–11]. As a result, DM signals are expected to correlate more strongly with low-redshift gravitational tracers. Beyond its application in the search for DM, the cross-correlation technique also holds broader potential for studying unresolved γ -ray source populations, offering insights into their redshift distribution and clustering properties.

Cross-correlations analyses have been performed between *Fermi*-LAT γ -ray sky maps and various tracers of the large-scale structure. These include weak gravitational lensing [12–15], the clustering of galaxies [16–22], galaxy clusters [23–27], and the lensing effect of the cosmic microwave background [28], which traces the large-scale distribution of matter across cosmological distances. Additional studies are presented in Refs. [7, 11, 29–31]). Positive cross-correlation signals have been found with galaxies at the level of 3.5σ [16–18] up to $8 - 10\sigma$ [22], galaxy clusters at 4.7σ [23], lensing of the cosmic microwave background at 3.2σ [28] and weak gravitational lensing at 5.3σ [15] and more recently at signal-to-noise ratio of 8.9 [32].

In this work, we explore the potential of the cross-correlation and autocorrelation techniques to detect a signal originating from subthreshold extragalactic source populations or from annihilation or decay of TeV-scale DM particles by using the CTAO extragalactic survey capabilities.

II. THE CHERENKOV TELESCOPE ARRAY

The CTAO will represent the next generation of the imaging atmospheric Cherenkov telescopes (IACTs), improving the technology of the current ground-based γ -ray detectors, such as H.E.S.S., MAGIC, and VERITAS. It is designed to surpass the sensitivity of current IACTs

by an order of magnitude in the TeV regime, to improve the angular resolution, as well as to enhance the surveying and monitoring capabilities. CTAO will consist of two IACT arrays: one located in the Southern hemisphere (CTAO-South) and the other in the Northern hemisphere (CTAO-North) [1].

The configuration will use three different types of telescope: large-, medium-, and small-sized telescopes (LST, MST, and SST, respectively), detecting photons over a broad energy range from 20 GeV up to 300 TeV. The LSTs will be most sensitive at lower energies, while the SSTs will cover the highest energies. The SSTs will be dedicated primarily to the observation of phenomena occurring within our Galaxy and will therefore be deployed only at the southern site from which the central/inner part of the Galaxy is most visible. LSTs and MSTs will instead be built both in the Northern and Southern locations and will focus on observations of extragalactic targets.

The telescopes will be equipped with cameras featuring wide fields of view: 4.3° for LSTs, $7.5-7.7^\circ$ for MSTs, and 8.8° for SSTs. A wide field of view is important for observing extended sources and diffuse emission, as well as for reducing the systematic errors of the measurements when requiring a uniform response over regions much larger than the telescope point-spread function (PSF). Several surveys covering large portions of the sky are planned as part of CTAO's observational strategy, and will be performed as Key Science Projects (KSPs) of the observatory early in the observatory's operation [1]. Thanks to its improved sensitivity and wider field of view, CTAO will surpass the surveying capabilities of existing IACTs. In particular, the extragalactic survey (detailed below) will be the largest survey ever undertaken by IACTs.

At the same time, CTAO will carry out a large number of traditional, dedicated pointings across the sky. The off-source data collected through these observations will cumulatively cover a substantial sky area with a significant average exposure. In this work we consider extragalactic data obtained from both observational strategies, as detailed in two subsections below.

A. The extragalactic survey

To study the auto and cross-correlation signals with gamma-rays, it is essential to use maps of large sky regions with approximately uniform coverage. Among CTAO Key Science Projects, the dedicated survey of the extragalactic sky—referred to as the extragalactic (EGAL) survey—presents the required characteristics.

As described in [1] (Sec. 8), the EGAL survey will cover a quarter of the sky, targeting the region defined by Galactic coordinates $-90^\circ < l < 90^\circ$ and $b > 5^\circ$. The survey is planned to reach an integral sensitivity ~ 6 mCrab above 125 GeV. One of its main objectives is to provide a high-resolution map of the extragalactic sky for photons with

energies at least up to 10 TeV and to compile a catalog of very high energy (VHE) extragalactic sources.

The EGAL observations are planned to be carried out over three years of observations, using the full CTAO Observatory (both the Southern and Northern arrays) to achieve nearly uniform sensitivity over the survey region. It is proposed that 15% of the sky will be covered by the Southern array using 400 h and 10% by the Northern array using 600 h, resulting in a total observation time of ~ 1000 h. The default observing strategy consists of a grid spaced by $\sim 3^\circ$ across the survey area, with each pointing observed for 0.51 h in the South and 1.11 h in the North. This configuration is expected to yield an effective exposure of approximately 3 h per point within the EGAL survey boundary.

B. Simulating survey observations

To simulate the planned EGAL survey observations, we use the software package CTOOLS,¹ developed for a scientific analysis of CTA data. We used the instrument response functions (IRFs) for the observatory,² labeled prod3b-v2, defining the so-called “omega” configuration. The respective IRF data files are publicly available at Ref. [33].

To enclose the region of the sky covered by the EGAL survey, we simulate observations using telescope-pointing directions corresponding to a HealPix realization with $\text{NSIDE} = 16$, resulting in 694 (3072) pointings within the EGAL survey bounds (over the whole sky). With this setup, the typical angular separation between adjacent pointings is $\sim 3.7^\circ$. For definiteness, the radius of the field of view (FoV) of each pointing is set to 6° . The pointing directions used for the simulation of the EGAL survey within the region of interest (ROI) are shown in Fig. 1.

The overlap of the FoV from different pointings contributes to a more uniform coverage of the survey’s ROI, minimizing the fluctuations in the exposure that inevitably occur when smaller FoV observations are used to cover a large survey area. Depending on the selected HealPix grid size (i.e., the values of the NSIDE parameter) and the pointing radius for individual observations, the overlap between nearby pointings can vary in extent. The overlaps between pointings are schematically shown in Fig. 2, where the upper panel presents a zoomed-in scheme of the pointing strategy from Fig. 1, and the bottom panel shows the simulated exposure map showing variations relative to the average exposure, highlighting the anisotropic pattern created by overlapping pointing regions.

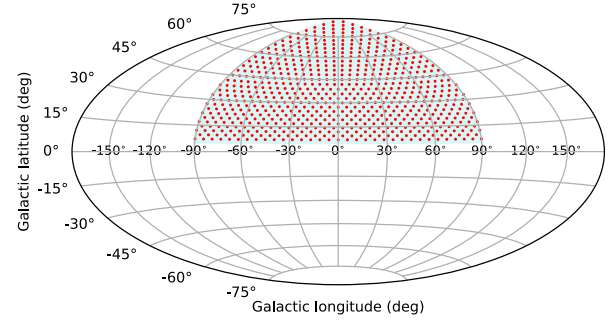


FIG. 1. Pointing directions for the region of the sky planned to be covered by the extragalactic survey. The pointing directions are shown using a HealPix pixelization with $\text{NSIDE} = 16$.

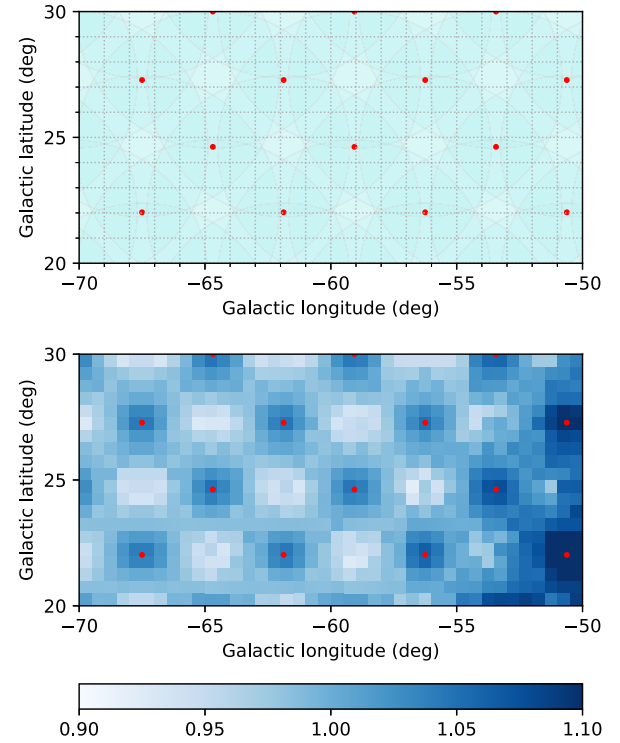


FIG. 2. Overlapping areas of different pointings. Top: A zoomed-in scheme of the pointing strategy shown in Fig. 1 (red), together with the overlapping fields of view of individual pointings (turquoise). Bottom: Simulated exposure map showing variations relative to the average exposure, highlighting the anisotropic pattern created by overlapping pointing regions.

The IRFs we use are optimized for observations under different zenith angles. For each pointing direction, we assign one of the three available IRF optimizations based on the minimum zenith angle at which that direction can be observed. To estimate the minimal zenith angle, we use the Earth coordinates of both array positions, i.e., $(l_N, b_N) = (17.8920, 28.7622)$ and $(l_S, b_S) = (79.4041, -24.6272)$ for the Northern and Southern array, respectively. The zenith angle is calculated as the angular distance between the array location and the pointing direction when they are

¹<http://CTA.irap.omp.eu/ctools/>.

²See <http://www.CTA-observatory.org/science/CTA-performance/> for more details.

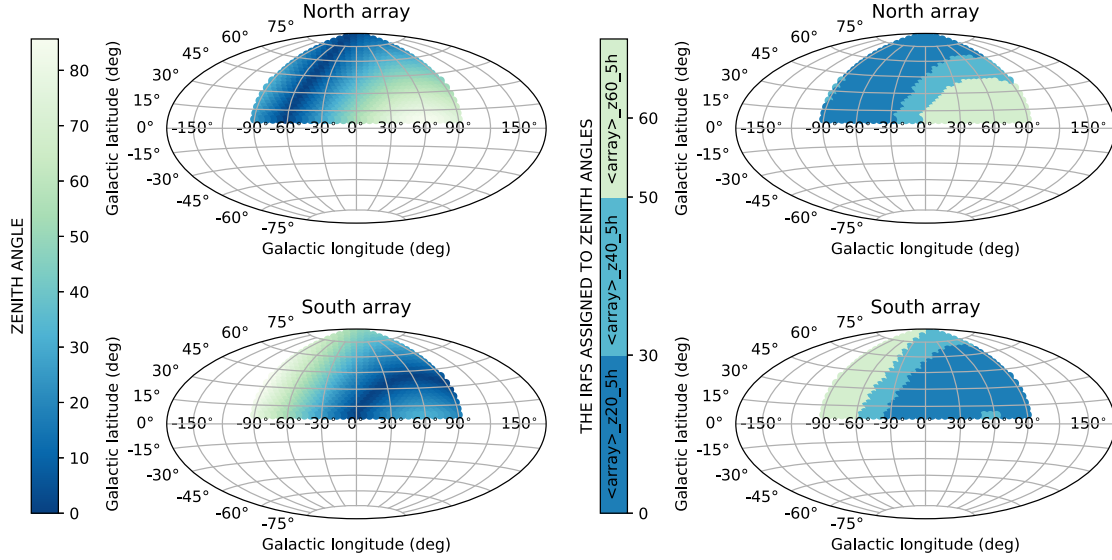


FIG. 3. Strategy for assigning instrument response functions to different sky regions within the planned extragalactic survey area, based on zenith angle.

closest. The resulting zenith angles for the Northern and Southern arrays are shown in the left panel of Fig. 3. The strategy for assigning the IRFs based on zenith angle is adapted from Ref. [34]. For zenith angles smaller than 30° , we use the IRF optimized for 20° angles; for zenith angles between 30° and 50° we use the IRF optimized for 40° angles, and for zenith angles larger than 50° we use IRF optimized for 60° angles, as shown in the right panel of Fig. 3. All IRFs used in this work are optimized for 5 hours of observation time, independently of the array or the zenith angle.

The observational strategy presented in Sec. 8 of Ref. [1] divides the extragalactic survey area into two parts: roughly $\sim 15\%$ of the sky to be observed using the Southern array and $\sim 10\%$ with the Northern array. However, the precise division of the survey observations between the two arrays has not yet been finalized. The planned strategy remains under discussion and is still subject to adjustments. For simplicity, we simulate observations of each pointing direction within the survey boundary using both Southern and Northern arrays with the corresponding zenith angle optimized IRFs. We use 1.11 hours and 0.51 hours per pointing direction with the Northern and Southern arrays, respectively. This strategy provides a reasonable approximation for simulating the survey when combined with zenith-angle-optimized IRFs, as regions observed at higher zenith angles will yield fewer detected events, and vice versa. As a result, the sky coverage of both arrays will be complementary, as illustrated in the right panel of Fig. 3.

The survey observations are simulated for energies between 30 GeV and 30 TeV, using 30 log-spaced energy bins to ensure proper sampling of the IRFs across the energy range. For the purpose of estimating the background

events and exposure over the whole extragalactic survey, we simulate the entire survey ROI, using coarser spatial binning of $1.0^\circ \times 1.0^\circ$ per pixel. For the sensitivity analysis, we instead simulate templates covering an ROI of $6.0^\circ \times 6.0^\circ$, using a finer spatial binning of $0.02^\circ \times 0.02^\circ$ per pixel.

C. Off-source data scenario

In addition to performing extensive surveys of the γ -ray sky, CTAO will be operated as an open, proposal-driven observatory. The observation time awarded to the wider research community can be added to the dedicated survey observation times of CTAO, and increasing the overall exposure that will be achieved over the first 10 years of telescope operation. An educated guess based on the performance of current IACTs can be made for CTAO observations performed through the proposal-driven program. In Ref. [35], the authors calculate the sky coverage and resulting observation time of the CTAO proposal program, assuming similarity with the actual operations of the MAGIC telescope. They extrapolate the 6.5 years of data obtained with the MAGIC array to 10 years of observation time using the two CTAO arrays. We adopt a similar approach to estimate the most optimistic case scenario for the observations of the extragalactic sky with CTAO, assuming an effective observation time of ~ 50 hours per point and the same sky coverage as in the EGAL survey, consistent with the findings in Ref. [35]. In the sensitivity analysis described below, we use a corresponding re-scaling factor to estimate the point source sensitivity of CTAO under this scenario. Details on how we derive the effective observation time for this scenario can be found in Appendix A.

III. THE CROSS-CORRELATION SIGNAL

The cross-correlation between a map of the UGRB, $\gamma(\hat{n})$, and fluctuations in galaxy number counts, $g(\hat{n})$, can be measured through the cross-correlation angular power spectrum (APS) [9–11]. In our analysis the field g will be the distribution of galaxies, which is a biased tracer of the underlying dark matter distribution. In the Limber approximation [36–38], the APS can be written as

$$C_{\ell}^{\gamma g} = \int \frac{d\chi}{\chi^2} W_{\gamma}(\chi) W_g(\chi) P_{\gamma g}(k = \ell/\chi, \chi), \quad (1)$$

where ℓ denotes the multipole, k the wavenumber, $\chi(z)$ is the radial comoving distance, related to redshift z in a flat cosmology through the relation $d\chi = cdz/H(z)$ with $H(z)$ being the expansion rate of the Universe. The term $P_{\gamma g}(k, z)$ represents the 3D cross-correlation power spectrum (PS), embedding the information on the fluctuations of the two correlated fields. The function $W_{i=\{\gamma, g\}}$ denotes the window functions of the γ -ray intensity field, γ , and the galaxy number counts fluctuations, g , respectively, and describe the redshift distribution of each field (see Appendix B for full details).

As astrophysical γ -ray sources we consider only BL Lacs, subdivided into the two subcategories of high-synchrotron-peaked (HSP) blazars and low and intermediate synchrotron-peaked (LISP) blazars [39]. We model the spectral energy distribution (SED) of these sources as power laws with index Γ and exponential cutoffs E_{cut} , whose specific parameter values are reported in Table I, together with the minimum and maximum luminosities of the population. Since we focus on CTAO and thus on energies above approximately 20 GeV, other typical γ -ray emitters like misaligned AGNs, star-forming galaxies and flat-spectrum radio quasars can be neglected, since their contribution are dominant at lower energies. Full details on the modeling of the astrophysical sources are provided in Appendix B, together with the expressions for their window functions and 3D cross-correlation power spectra, obtained within the halo model formalism.

For the DM γ -ray emission, we consider both annihilating and decaying DM. In the case of annihilation, the relevant particle physics parameters are the DM mass m_{χ} and the thermally-averaged annihilation cross section $\langle\sigma v\rangle$.

TABLE I. Parameters and uncertainties for the power-law with exponential cut-off SED models of HSP and LISP populations, including the minimum and maximum luminosity, adopted from Table 1 of Ref. [39].

	Γ	E_{cut} [GeV]	L_{min} [erg/s]	L_{max} [erg/s]
HSP	1.86	910^{+1100}_{-450}	10^{38}	3×10^{45}
LISP	2.08	37^{+85}_{-20}	10^{38}	3×10^{46}

For decaying DM, the key parameters are the mass and the lifetime τ . These will be the parameters for which we provide the forecast sensitivity in the cross-correlation analysis. The expressions for the window functions and the 3D cross-correlation power spectra are reported in Appendix B. Here we recall that the window functions are proportional to $\langle\sigma v\rangle$ for annihilating DM and inversely proportional to τ in the case of decaying DM. A few examples of the redshift behavior of the γ -ray window functions for DM annihilation are shown in Fig. 4, for two photon energies and two DM masses. For comparison, we also show the window functions of unresolved LISP and HSP astrophysical source classes, assuming an observation time of 3 hours. The observation time affects the window functions of astrophysical sources, since it determines the minimum luminosity of the sources that can be resolved by the instrument (and therefore do not contribute to the unresolved component consider in our analysis). The case for 50 hours of observation is shown in Fig. 15 in Appendix B.

For the present analysis, which primarily focuses on the sensitivity to a DM signal, we adopt as galaxy catalogs the Two Micron All-Sky Survey (2 MASS) [40,41] and the 2 MASS Redshift Survey (2MRS) [42,43] catalogs, since they cover the low redshift range $z \lesssim 0.1$ where most of the DM γ -ray emission originates. Astrophysical sources, on the other hand, have a peak in emission at redshifts $z \sim 0.3\text{--}0.4$ at $E \sim 50$ GeV, and $z \sim 0.1\text{--}0.2$ at $E \gtrsim 1$ TeV as shown in Fig. 4 and in Fig. 15 in Appendix B. The latter range is a consequence of strong absorption of γ -rays above ~ 1 TeV due to interactions with the extragalactic background light (EBL), which limits the γ -ray horizon to $z \sim 0.1\text{--}0.2$ independently of the type of source.

The present analysis is thus optimized for DM searches, while for astrophysical sources it will be beneficial to consider galaxy catalogs covering higher redshifts, up to $z \sim 0.4\text{--}0.5$. We leave this extension to future studies. Additional details on the window functions of the galaxy catalogs can be found in Appendix B, together with the 3D cross-correlation power spectra. The modeling of galaxy clustering within halos is based on the halo occupation distribution (HOD) formalism, which is also outlined in Appendix B.

Let us note that in our analysis, the window function $W_g(z)$ of the galaxy catalogs is normalized to unity; $\int dz W_g(z) = 1$. For the γ -ray window function $W_{\gamma}(\chi)$, we define it such that the (average) intensity energy spectrum is obtained as: $I_{\gamma}(E) = \int d\chi W_{\gamma}(E, \chi)$.

The cross-correlation APS signal has a variance $(\Delta C_{\ell}^{\gamma g})^2$ given by

$$\begin{aligned} & (\Delta C_{\ell}^{\gamma g})^2 (2\ell + 1) f_{\text{sky}} \\ &= (C_{\ell}^{\gamma g})^2 + \left[C_{\ell}^{\gamma\gamma} + \frac{C_N}{(B_{\ell})^2} \right] (C_{\ell}^{gg} + C_{N_g}), \end{aligned} \quad (2)$$

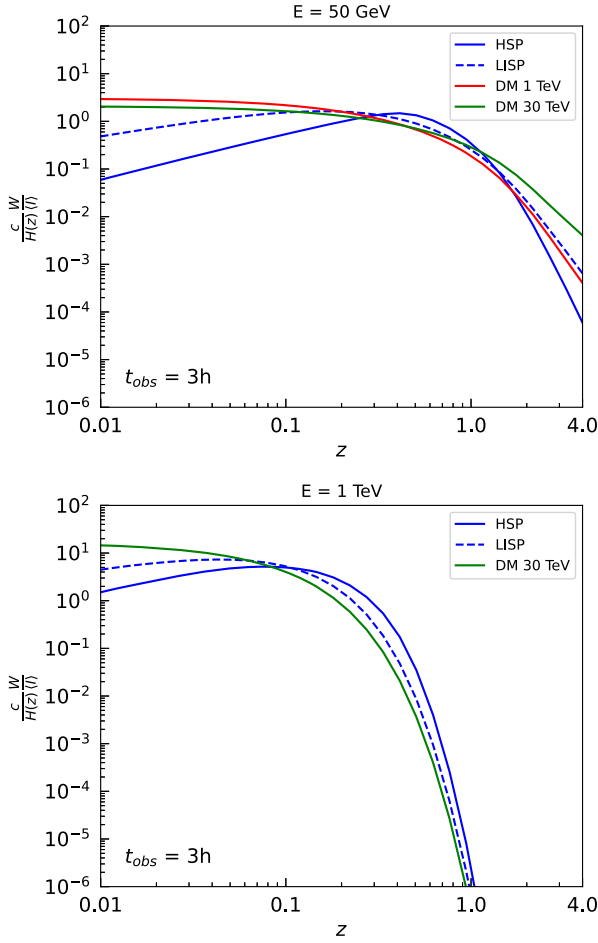


FIG. 4. The γ -ray window functions, normalized to the mean γ -ray intensity and divided by the Hubble parameter, as a function of redshift. Blue solid and dashed lines show HSP and LISP astrophysical sources for 3 hours of observation, respectively. Red and green lines correspond to DM annihilation into $b\bar{b}$, $\langle\sigma v\rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$ for DM masses of 1 TeV (red) and 30 TeV (green). Upper panel: $E_\gamma = 50 \text{ GeV}$; lower panel: $E_\gamma = 1 \text{ TeV}$ (only 30 TeV DM case shown).

where $C_\ell^{\gamma\gamma}$ and C_ℓ^{gg} are the γ -ray and galaxy autocorrelation APS, C_N is the photon noise, C_{N_g} the galaxy catalog noise, and B_ℓ denotes the beam correction accounting for the finite angular resolution of the γ -ray instrument. No beam correction is applied to the galaxy term, since the angular resolution of galaxy catalogs (typically of the order of arcseconds) is negligible for the purposes of this analysis. The parameter f_{sky} denotes the fraction of the sky covered by the analysis. All the relevant terms in Eq. (2) are outlined in Appendix B.

For completeness, in the following analysis we will also use the γ -ray autocorrelation APS, which is a measure of the intrinsic anisotropy of the γ -ray emission. It reads

$$C_\ell^{\gamma\gamma} = \int \frac{d\chi}{\chi^2} W_\gamma^2(\chi) P_{\gamma\gamma}(k = \ell/\chi, \chi), \quad (3)$$

where $P_{\gamma\gamma}$ is the 3D power spectrum of the γ -ray emission. The corresponding variance on $C_\ell^{\gamma\gamma}$ is

$$(\Delta C_\ell^{\gamma\gamma})^2 = \frac{2}{(2\ell + 1)f_{\text{sky}}} \left[C_\ell^{\gamma\gamma} + \frac{C_N}{(B_\ell)^2} \right]^2. \quad (4)$$

IV. SENSITIVITY FORECAST

A. Astrophysical sources

As a first step, we determine whether the anisotropy and the cross-correlation signals from unresolved astrophysical sources are detectable. As a statistical figure of merit, we adopt the signal-to-noise ratio (SNR), defined, for the cross-correlation, as

$$\text{SNR} = \sqrt{\sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma g}}{\Delta C_{\ell,i}^{\gamma g}} \right)^2}, \quad (5)$$

where the sum runs over all the multipoles ℓ considered in the analysis and all energy bins i . For the autocorrelation, the formula is analogous but using $C_{\ell,i}^{\gamma\gamma}$ and $\Delta C_{\ell,i}^{\gamma\gamma}$. We start the sum over multipoles from $\ell = 10$ in order to avoid the largest angular scales, where effects due to the fluctuations of the galactic foreground can sizably affect $C_\ell^{\gamma\gamma}$, thus masking the autocorrelation APS signal at low multipoles and contributing to the variance also in the cross-correlation case [see Eq. (2)].

The results for 3 hours and 50 hours of observation time are shown in Table II, for both LISP and HSP source classes, and for both the autocorrelation and cross-correlation APS with the 2 MASS galaxy catalog. We find that a purely astrophysical signal from unresolved sources can reach at most (for 50 hours of observation) only a marginal evidence of $\text{SNR} \simeq 0.9$ in the autocorrelation, and about 0.16 in the cross-correlation case. These values of SNR correspond to the HSP blazars, whose hard spectra are better suited to the CTAO energy range. The SNR is even smaller for LISPs, whose softer spectra result in a poorer overlap with the CTAO sensitivity.

Note that this is not necessarily a negative result; it simply reflects that a large fraction of blazars will be resolved, thereby reducing the auto and cross-correlation APS of the unresolved population. In this regime, it may be actually convenient to perform auto and cross-correlation APS analyses of the catalog of resolved sources. This result, however, strongly depends on the assumed point-source sensitivity of the detector, which for CTAO remains subject to considerable uncertainty (as discussed in Sec. II). This is due to the fact the final configuration and specifications of CTAO have not yet been fully settled. For example, the point-source sensitivity differs significantly between CTAO-North and CTAO-South [1], and at the

TABLE II. The predicted signal-to-noise ratios for 3-hours and 50-hours observations of LISP and HSP source classes, obtained for both γ -rays autocorrelation and cross-correlation with the 2 MASS galaxy catalog.

SNR for 3 hrs	Autocorrelation	Cross with 2 MASS
LISP	0.042	8.2×10^{-3}
HSP	0.31	0.1

SNR for 50 hrs	Autocorrelation	Cross with 2 MASS
LISP	0.18	0.022
HSP	0.9	0.16

TABLE III. Same as Table II but for 50 hrs-observations, assuming a source detection threshold approximately three times worse than in Table II. This illustrates the strong dependence of the autocorrelation signal on the point source detection threshold.

SNR 50 hrs low res	Autocorrelation	Cross with 2 MASS
LISP	0.7	0.033
HSP	5.1	0.41

moment the relative share of North and South array observations to the survey is still undecided.

To better illustrate this point, in Table III we calculate the SNR for 50 hrs of observation, but assuming a degraded point-source detection capability. For definiteness, we take the sensitivity corresponding to 3 hrs of observations, which is approximately a factor of 3 worse than the nominal 50 hrs sensitivity (Note that, since we are in a strongly background-dominated regime, the uncertainty ΔC_ℓ does not depend on the C_ℓ itself, but only on the background and the noise, as discussed in Appendix B). The result shows that the autocorrelation SNR for HSP

indeed increases significantly, reaching $\text{SNR} \simeq 5$. The detectability of the cross-correlation APS signal also improves, but not so strongly as in the autocorrelation case. This is because the cross-correlation signal is less sensitive to the point-source detection threshold. As detailed in the Appendixes, the point-source sensitivity impacts the cross-correlation signal only through the window function, whereas it affects both the window function and the 3D power spectrum in the autocorrelation case.

Regarding LISPs, on the other hand, the SNR remains very low in both cases, reaching at most 0.7 for the autocorrelation and 0.03 for the cross-correlation. This is expected, because LISPs have a SED cutoff around 40 GeV resulting in minimal emission in the energy range where CTAO is most sensitive.

B. Dark matter

We now turn to assessing the CTAO sensitivity to an auto or cross-correlation APS originated from DM annihilation or decay. To this end, we compare the null hypothesis, where only the contribution from astrophysical sources is present, with the alternative hypothesis scenario in which a DM γ -ray component is added on top of the astrophysical signal. From this comparison, we derive the minimum DM signal required to be statistically distinguishable from the purely astrophysical case.

For cross-correlation, we adopt the following test statistic:

$$\Delta\chi^2 = \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}} + \text{DM}g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}} + \text{DM}g}} \right)^2 - \sum_{\ell,i} \left(\frac{C_{\ell,i}^{\gamma_{\text{astro}}g}}{\Delta C_{\ell,i}^{\gamma_{\text{astro}}g}} \right)^2, \quad (6)$$

where $C_{\ell,i}^{\gamma_{\text{astro}}g}$ denotes the cross-correlation signal due to the presence of only astrophysical sources, while $C_{\ell,i}^{\gamma_{\text{astro}} + \text{DM}g}$ refers to the case where a DM-induced γ -ray emission is present. The corresponding standard deviations are $\Delta C_{\ell,i}^{\gamma_{\text{astro}}g}$

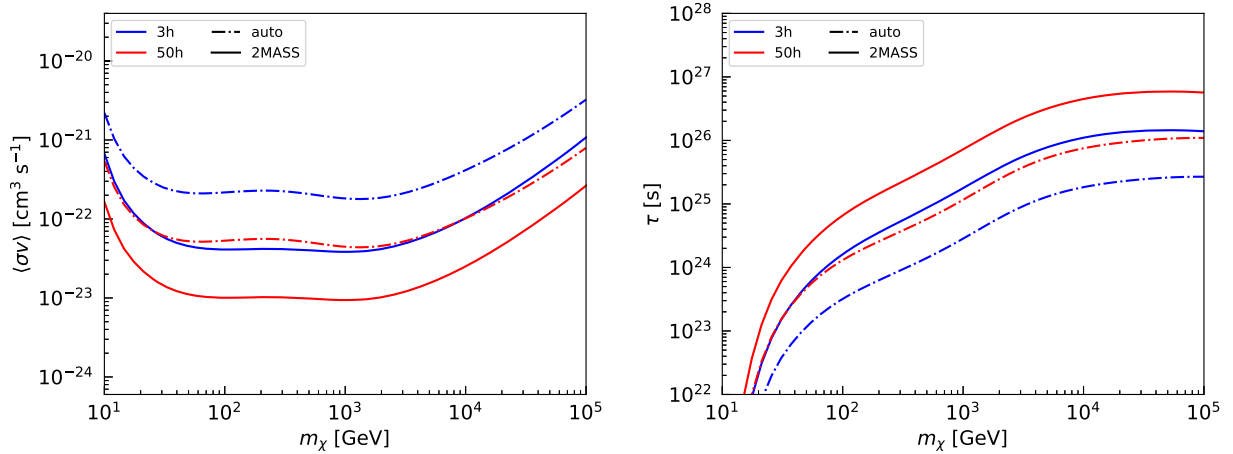


FIG. 5. Left: Predicted upper bounds on the dark matter cross section $\langle\sigma v\rangle$, for annihilation into $b\bar{b}$, based on 3 hours and 50 hours of observations. The curves represent the bounds from γ -ray autocorrelation (dot-dashed) and cross-correlation with the 2 MASS galaxy catalog (solid). Right: Same as the left panel, but for the lower bound on the DM particle lifetime τ .

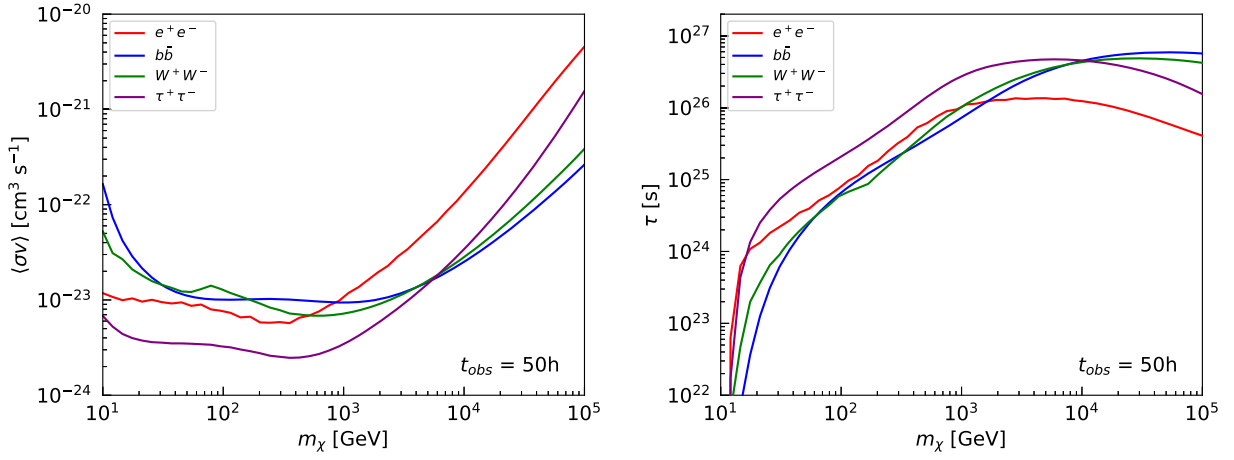


FIG. 6. Left: Predicted upper bounds on the dark matter cross section $\langle\sigma v\rangle$ from the cross-correlation with 2 MASS galaxies, based on 50 hours of observation, for various dark matter annihilation final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. Right: Same as the left panel, but for the lower bound on the DM particle lifetime τ .

and $\Delta C_{\ell,i}^{\gamma_{\text{astro}}+\text{DM}^g}$, respectively. Since the DM signal depends on the DM mass m_χ and scales linearly with the annihilation rate $\langle\sigma v\rangle$ (or inversely with the lifetime τ in the case of decay), as outlined in Appendix B, we derive bounds on the annihilation (or decay) rate by scanning over the DM mass m_χ . We then derive the 2σ bounds on the annihilation (or decay) rate by requiring $\Delta\chi^2 = 4$. A similar test statistic and procedure is employed for the autocorrelation case.

Figure 5 shows the sensitivity to the DM annihilation and decay in the $b\bar{b}$ annihilation channel, derived from the autocorrelation and cross-correlation APS with the 2 MASS catalog, for 3 hrs and 5 hrs of CTAO observations. The 5 hrs case yields approximately a factor of 4

improvement in sensitivity compared to the 3 hrs case, while the cross-correlation outperforms the autocorrelation by approximately a factor of 5. This is expected since in the autocorrelation the DM signal competes with strong blazar autocorrelation, while in the cross-correlation the blazar contribution is much lower, yielding a more favorable signal-to-background ratio. Sensitivities obtained from the cross-correlation with the 2MRS catalog are very similar and are shown Appendix B. Note that in this case, the uncertainty on the point-source detection threshold does not play a significant role, since the DM signal is always diffuse.

Figure 6 shows the sensitivities of the cross-correlation technique for different annihilation channels, while Fig. 7

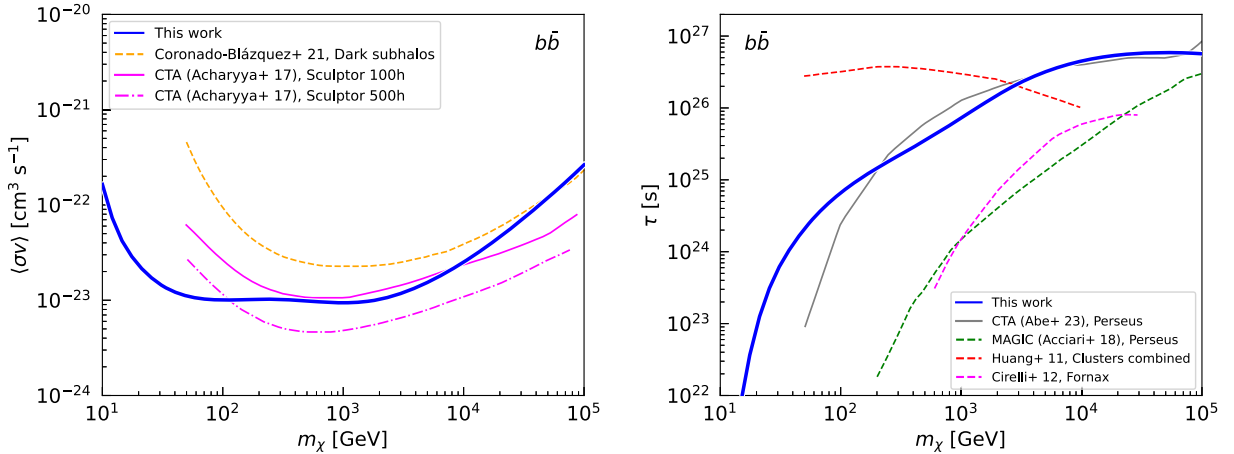


FIG. 7. Left: Comparison between the forecast on the upper bounds on the annihilation cross section $\langle\sigma v\rangle$ obtained in this work with other bounds from the literature, for the $b\bar{b}$ annihilation channels: bounds from nonobservation of dark subhalos (dashed yellow) [35], from Sculptor dwarf galaxy [44] with 100 hours (solid magenta) and with 500 hours (dot-dashed magenta) of observations with CTAO. Right: Comparison between the forecast on the lower bounds on the DM particle lifetime τ obtained in this work with other bounds from the literature, for the $b\bar{b}$ decay channel; bound from Perseus with CTAO [45] and MAGIC [46], from galaxy clusters [47] and from Fornax [48].

provides a comparison with existing results in the literature for alternative DM search strategies. In particular, we show bounds from observations of dwarf galaxies [44] and dark subhalos [35] for DM annihilation, and from galaxy clusters for decaying DM [45–48]. Our results show that cross-correlation technique is competitive, yielding constraints comparable to those from other methodologies. Note that the figure does not include the sensitivity from the Galactic center DM searches, which can typically probe lower values of $\langle\sigma v\rangle$, but are strongly dependent on the assumptions regarding the DM profile near the Galactic center.

In Appendix C we show a comprehensive set of plots for different hours of observations and DM channels.

V. SUMMARY AND CONCLUSIONS

The CTAO will be the next generation of ground-based γ -ray Cherenkov detectors, offering a unique opportunity to study high-energy phenomena in the TeV range. In addition to targeted observations, CTAO is expected to carry out large-area sky surveys, making it particularly well-suited for anisotropy-based techniques such as autocorrelation and cross-correlation analyses. In this work, we have investigated the sensitivity of future CTAO data to constrain the properties of unresolved blazars and DM signals using these methods.

We found, in particular, that the autocorrelation APS method is better suited for probing the signal from unresolved blazars, with its sensitivity strongly dependent on the effective point-source detection threshold. On the other hand, the cross-correlation APS technique, when used in combination with galaxy catalogs, offers very promising sensitivity to potential DM annihilation or decay signal, reaching values around $10^{-23} \text{ cm}^{-3} \text{ s}^{-1}$ for annihilation cross section $\langle\sigma v\rangle$ and 10^{27} s for decay. These sensitivities are comparable to and complementary with, those obtained from existing strategies such as observations of dwarf spheroidal galaxies or galaxy clusters.

In the future, these results can be further improved with upcoming, deeper, and more densely sampled galaxy catalogs, such as those from the European Space Agency’s *Euclid* satellite mission [49], as well as by incorporating cross-correlations with other large-scale-structure tracers, such as weak lensing.

ACKNOWLEDGMENTS

E. P. is grateful for the hospitality of Perimeter Institute where part of this work was carried out. Research at Perimeter Institute is supported in part by the Government of Canada through the Department of Innovation, Science and Economic Development and by

the Province of Ontario through the Ministry of Colleges and Universities. A. A. acknowledges support from Generalitat Valenciana through the ‘GenT program’, Ref.: CIDEAGENT/2020/055 (PI: B. Zaldivar). A. C. and N. F. acknowledge support from the Research grant TAsP (Theoretical Astroparticle Physics) funded by INFN. A. C. acknowledges support from: Departments of Excellence grant awarded by the Italian Ministry of University and Research (MUR), and Research grant “Addressing systematic uncertainties in searches for dark matter”, Grants No. 2022F2843L and No. CUP D53D2300258 0006 funded by MUR. S. C. acknowledges support from MUR, PRIN 2022 ‘EXSKALIBUR—Euclid-Cross-SKA: Likelihood Inference Building for Universe’s Research’, Grants No. 20222BBYB9 and No. CUP D53D2300252 0006, and from the European Union—Next Generation EU. N. F. is supported by the European Union—Next Generation EU and by MUR via the PRIN 2022 project n. 20228WHTYC. G. Z. acknowledges Slovenian Research Agency Grants No. P1-0031, No. I0-0033, and No. J1-60014 and the Young Researcher program, Slovenia. The authors gratefully acknowledge the computer resources at Artemisa and the technical support provided by the Instituto de Física Corpuscular, IFIC(CSIC-UV), European Regional Development Fund Operative Programme of Comunitat Valenciana and Consejo Superior de Investigaciones Científicas. Artemisa is cofunded by the European Union through the 2014-2020 ERDF Operative Programme of Comunitat Valenciana, Project No. IDIFEDER/2018/048. This work was conducted in the context of the CTAO ‘Dark Matter and Exotic Physics’ Working Group.

DATA AVAILABILITY

The data that support the findings of this article are openly available on [50].

APPENDIX A: OFF-SOURCE DATA SCENARIO

As described in the main text, we follow the strategy outlined in Ref. [35] to extrapolate MAGIC observations to CTAO. The distribution of MAGIC pointings, that we consider when extrapolating the observation time and pointing directions of future CTAO measurements across the sky, is shown in Fig. 8, together with one realization of the predicted CTAO distributions of the same quantities, i.e. the Galactic latitude, longitude, and the observation time per pointing. The distribution of the cumulative pointing time over all pointings, obtained from 1000 different realizations of future CTAO observations, is shown in Fig. 9.

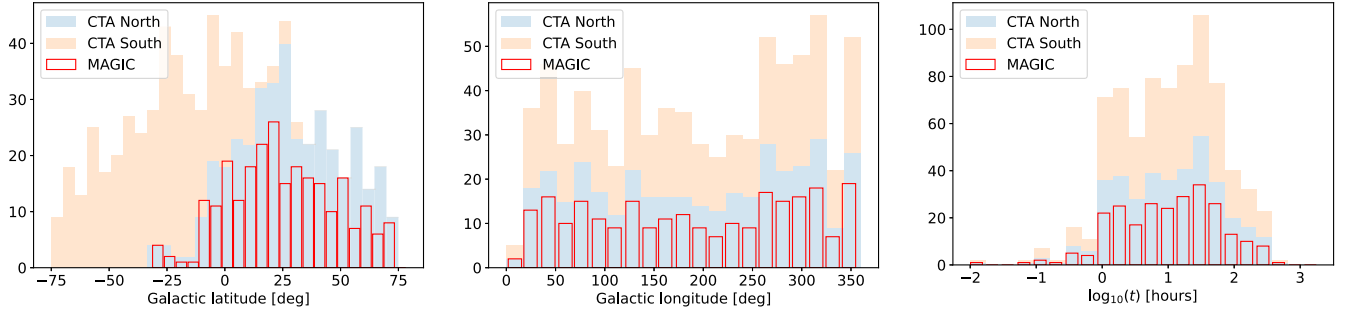


FIG. 8. Distributions of Galactic latitude (left), longitude (center), and observation time (right) for one realization of the predicted CTAO pointings over 10 years of observations. The stacked histograms show contributions from the Northern (blue) and Southern (orange) arrays. For comparison, the red outlined histograms indicate the corresponding distributions from 6.5 years of MAGIC observations.

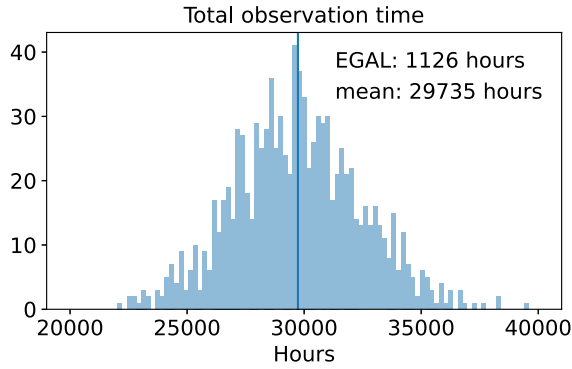


FIG. 9. Distribution of the total observation time from 1000 realizations of predicted CTAO pointings across the whole sky using both arrays. The blue vertical line marks the mean total observation time. The total time in the most optimistic scenario is about 25 times higher than that of the EGAL survey, effectively translating the nominal 3 hours per pointing into ~ 50 hours per sky location in this scenario.

APPENDIX B: FORMALISM AND DETAILS OF THE MODELING

1. Angular power spectra

The angular power spectra for the γ -ray autocorrelation and for the cross-correlation between the unresolved γ -ray emission and a galaxy catalog g can be expressed as

$$C_{\ell}^{\gamma\gamma} = \int \frac{d\chi}{\chi^2} W_{\gamma}^2(\chi) P_{\gamma\gamma}(k = \ell/\chi, \chi), \quad (\text{B1})$$

$$C_{\ell}^{\gamma g} = \int \frac{d\chi}{\chi^2} W_{\gamma}(\chi) W_g(\chi) P_{\gamma g}(k = \ell/\chi, \chi), \quad (\text{B2})$$

where ℓ denotes the multipole, χ is the comoving distance, $W_{\gamma}(\chi)$ and $W_g(\chi)$ are the window functions of the γ -ray emission and of the galaxy catalog, respectively, that is the functions describing how the observables are distributed in redshift. $P_{\gamma\gamma}$ and $P_{\gamma g}$ are the 3D power spectra of the auto and cross-correlation, respectively, which quantify the

fluctuations of the corresponding observables. All these components are discussed in the following subsections.

$C_{\ell}^{\gamma\gamma}$ and $C_{\ell}^{\gamma g}$ are both energy-dependent. In our approach, we compute the correlation functions within the specific energy bins, for which the detector sensitivity is available. CTAO sensitivities are provided for 19 energy bins spanning the range (5 GeV, 501 TeV). Due to the sensitivity at low energies, we restrict our analysis to 16 bins, logarithmically spaced within the 30 GeV–30 TeV range. For visualization purposes, the energy-dependent quantities are further coadded into 3 broader bins, as summarized in Table IV. These low-, medium-, and high-energy bins correspond to the CTAO bin numbers 2-7, 8-12, and 13-18, respectively. However, when computing the DM constraints, we make sure of all 19 energy bins available for CTAO (bins 1 to 19), thereby covering the full energy range from 5 GeV to 501 TeV.

The variance of the autocorrelation APS, in the Gaussian approximation, is given by Eq. (4).

The sky fraction observed by CTAO is $f_{\text{sky}} \sim 0.25$ and it is considered energy-independent in our analysis. The beam window functions B_{ℓ} account for the smearing of the angular signal due to the finite angular resolution of the instrument and can be defined in terms of the point spread function (PSF) as

$$B_{\ell}(E) = \int d\theta \text{PSF}(\theta, E) \sin \theta P_{\ell}(\cos \theta), \quad (\text{B3})$$

where $P_{\ell}(\cos \theta)$ are the Legendre polynomials of order ℓ . The point-spread function $\text{PSF}(\theta, E)$ has been obtained

TABLE IV. Coadded energy bins used in the analysis for display purposes.

Bin	$E_{\min}$	$E_{\max}$
Low	8 GeV	126 GeV
Mid	126 GeV	1.3 TeV
High	1.3 TeV	31.6 TeV

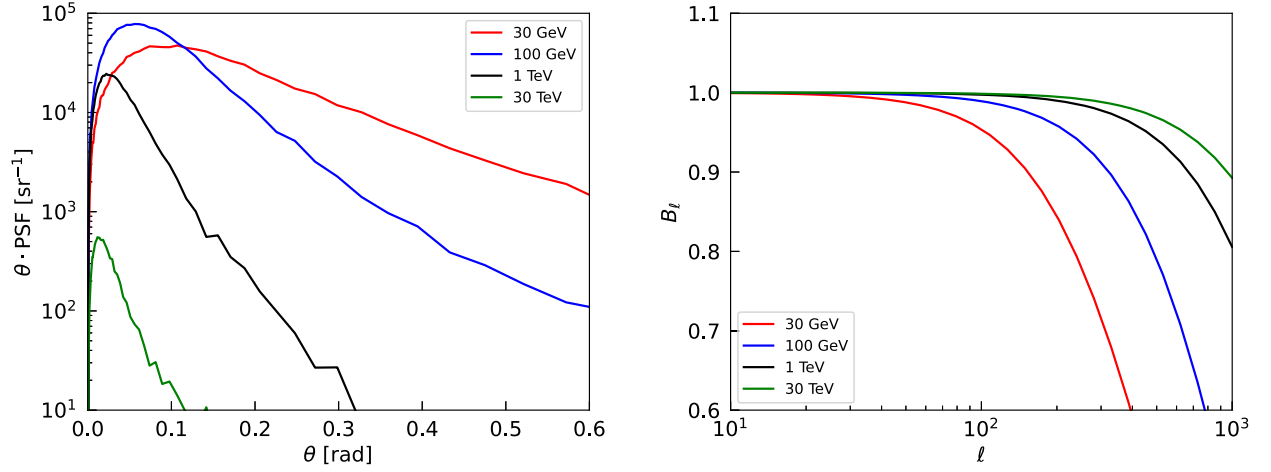


FIG. 10. Left: Point spread functions of CTAO, as a function of angle for different photon energies. Right: The corresponding CTAO beam window functions as a function of multipole.

through a Monte Carlo simulation of the CTAO detector, and is energy-dependent. Figure 10 shows the $\text{PSF}(\theta, E)$ as a function of the angle θ (left panel) and the corresponding beam window function as a function of the multipole ℓ (right panel) for different energies.

The beam window function in a specific energy bin is then obtained by averaging over the energy-dependent intensity flux dI/dE :

$$\langle B_\ell \rangle = \frac{1}{N} \int_{\text{bin}} dE B_\ell(E) \frac{dI}{dE}, \quad (\text{B4})$$

where $N = \int_{\text{bin}} dE dI/dE$. For definiteness, we use $dI/dE \propto E^{-2.3}$, which appropriately matches the observed behavior of the unresolved γ -ray intensity as measured by *Fermi*-LAT.

The photon noise C_N is also energy-dependent and is determined as

$$C_N = \frac{4\pi f_{\text{sky}}}{N_\gamma} \left(1 + \frac{N_{\text{CR}}}{N_\gamma} \right) \langle I \rangle^2, \quad (\text{B5})$$

where N_{CR} represents the cosmic-ray irreducible background, N_γ is the number of γ -ray events, and $\langle I \rangle$ denotes the mean γ -ray intensity in the specific energy bin, which is determined from the window functions as: $\langle I \rangle = \int d\chi W(\chi)$. N_γ is therefore related to $\langle I \rangle$ via:

$$N_\gamma = 4\pi f_{\text{sky}} \mathcal{E} \langle I \rangle, \quad (\text{B6})$$

where $\mathcal{E}$ represents the CTAO exposure in the given energy bin. The exposure and the counts associated with the cosmic-ray background are derived via Monte Carlo simulations of the CTAO array. The simulation has been performed for 1.62 h of observation time, and the results were then rescaled to obtain the exposure for 3 hrs and 50 h. Figure 11 displays the estimated cosmic-ray backgrounds

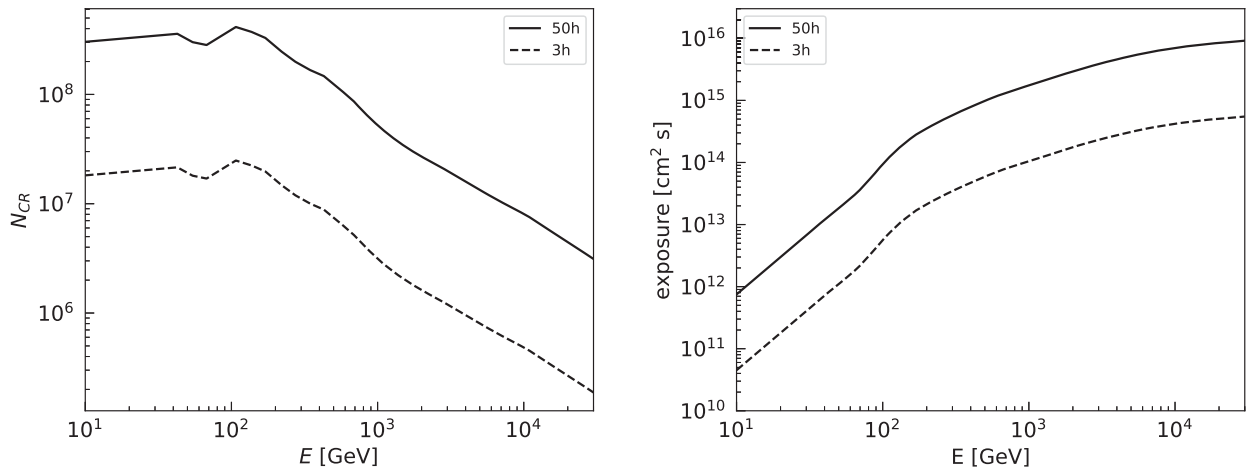


FIG. 11. Left: Cosmic-ray background N_{CR} as a function of energy. Right: CTAO exposure as a function of energy.

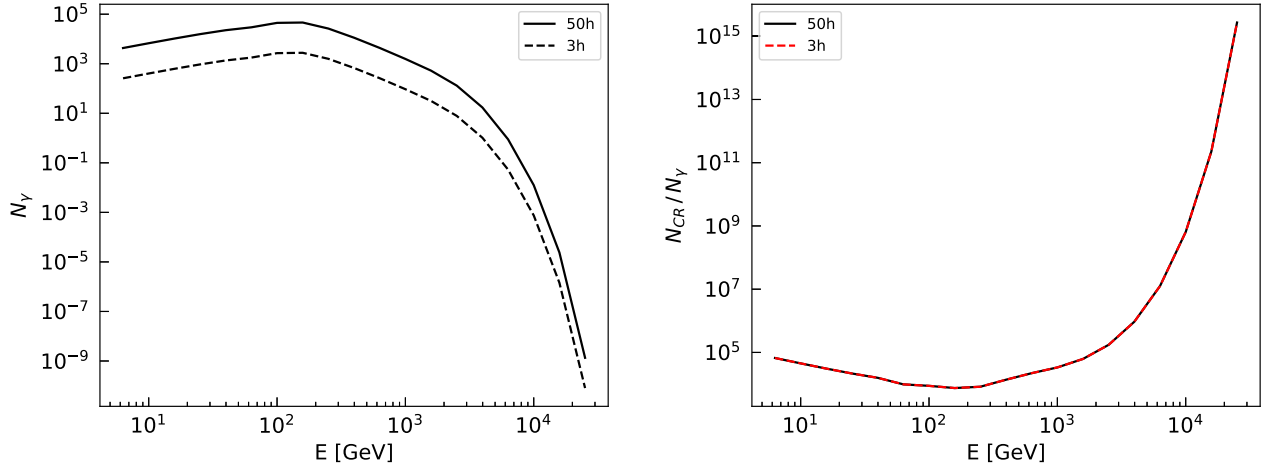


FIG. 12. Left: Number of expected γ -ray events N_γ in the CTAO survey as a function of energy. Right: Ratio of cosmic-ray events to expected γ -ray events (N_{CR}/N_γ) after quality cuts in the CTAO survey. The ratio worsens above 1 TeV due to significant γ -ray suppression from EBL attenuation. N_γ is determined for the HSP plus LISP source classes.

N_{CR} (left panel) and exposures $\mathcal{E}$ (right panel) as a function of the energy.

Figure 12 shows the number of γ -ray events N_γ and the ratio N_{CR}/N_γ as a function of energy. Since $N_{CR}/N_\gamma \gg 1$ in all the energy bins considered in our analysis, the noise can be approximated as

$$C_N \simeq 4\pi f_{\text{sky}} \frac{N_{CR}}{N_\gamma^2} \langle I \rangle^2. \quad (\text{B7})$$

By substituting the expression of N_γ in Eq. (B7), it becomes evident that C_N can be expressed only in terms of detector properties and configuration:

$$C_N \simeq \frac{N_{CR}}{4\pi f_{\text{sky}} \mathcal{E}^2}. \quad (\text{B8})$$

The noise C_N is shown explicitly in Fig. 21, together with a selection of auto and cross-correlation energy spectra.

In the case of cross-correlations, the variance is instead given by

$$(\Delta C_\ell^{gg})^2 = \frac{1}{(2\ell + 1)f_{\text{sky}}} \left[(C_\ell^{gg})^2 + \left(C_\ell^{gg} + \frac{C_N}{B_\ell^2} \right) (C_\ell^{gg} + C_{N_g}) \right], \quad (\text{B9})$$

where C_ℓ^{gg} is the galaxy autocorrelation, C_N is the same as in the autocorrelation case, C_{N_g} is the galaxy catalog noise. There is no beam correction in the galaxy term, as the angular resolution on the galaxy catalog is on the order of arcseconds, making it negligible in this analysis. The galaxy noise is given by

$$C_{N_g} = \frac{4\pi f_{\text{sky},g}}{N_g}, \quad (\text{B10})$$

where $f_{\text{sky},g}$ represents the fraction of the sky covered by the catalog, and N_g is the number of galaxies in the catalog. Figure 21 shows, the quantity $\sqrt{C_N C_{N_g}}$ as a representative level for the cross-correlation noise, which approximately corresponds to the noise expected in a noise-dominated regime.

2. Window functions

a. Gamma-ray astrophysical sources

The window function for the γ -ray emission of unresolved astrophysical sources can be written as

$$W(E, \chi) = \frac{dW}{dE d\chi} = \frac{d_L^2(z)}{(1+z)^2} \int_{L_{\min}}^{L_{\text{th}}(z)} dL \Phi(L, z) \times \frac{dF}{dE}(E, L, z) \exp[-\tau(E, z)], \quad (\text{B11})$$

where $d_L(z)$ is the luminosity distance, $\Phi(L, z)$ is the γ -ray luminosity function of the source population. In the present analysis we use the luminosity functions of HSP and LISP as reported in [39]. Note that the above window function, which is used in Eqs. (B1)–(B2) is differential in comoving distance, χ . The analogous quantity differential in z , would be $W(E, z) = \frac{dW}{dE dz} = \frac{dW}{dE d\chi} \frac{d\chi}{dz} = W(E, \chi) \frac{d\chi}{dz}$ where $\frac{d\chi}{dz} = \frac{c}{H(z)}$. Also note that the above window is differential in energy. In Eqs. (B1)–(B2), however, the corresponding quantity must be integrated over the energy bin considered. L is the γ -ray luminosity of the source, conventionally integrated over the interval (0.1, 100) GeV:

$$L = \int_{0.1 \text{ GeV}}^{100 \text{ GeV}} dE_r \frac{dL}{dE_r}, \quad (\text{B12})$$

where E_r is the source rest-frame energy, related to the observed energy by $E_r = (1+z)E$, and dF/dE is the

spectral energy distribution (SED) of the sources, which we model as a power law with index Γ and an exponential cutoff,

$$\frac{dF}{dE} = A(L, z) E^{-\Gamma} \exp(-E/E_{\text{cut}}). \quad (\text{B13})$$

The values of Γ and E_{cut} for the two classes of astrophysical sources adopted in our models are reported in Table I, together with the minimum luminosity of the source classes L_{min} . The parameter A , on the other hand, can be related to the luminosity L , by observing that [see Eq. (22) in [51]]

$$\begin{aligned} L &= \frac{4\pi d_L^2(z)}{(1+z)} \int_{0.1 \text{ GeV}}^{100 \text{ GeV}} dE_r E_r \frac{dF}{dE_r} \\ &= 4\pi d_L^2(z) \int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E \frac{dF}{dE}, \end{aligned} \quad (\text{B14})$$

and therefore

$$\begin{aligned} A(L, z) &= \frac{L}{4\pi d_L(z)^2} \left[\int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E^{(-\Gamma+1)} \exp(-E/E_{\text{cut}}) \right]^{-1}. \end{aligned} \quad (\text{B15})$$

The last term in Eq. (B11) models the γ -ray absorption on EBL. For the optical depth $\tau(E, z)$, we adopt the results from Ref. [52]. The maximal luminosity $L_{\text{th}}(z)$ in Eq. (B11), which refers to the threshold of detectability of resolved sources by the detector, depends on the CTAO sensitivity $F_{\text{sens}}^{\text{CTA}}(z)$ to point sources in the energy range of operation, which in our analysis is defined as (30 GeV, 30 TeV)

$$\begin{aligned} F_{\text{sens}}^{\text{CTA}}(z) &= \int_{30 \text{ GeV}}^{30 \text{ TeV}} dE \frac{dF}{dE} \exp[-\tau(E, z)] \\ &= A_{\text{th}}(z) \int_{30 \text{ GeV}}^{30 \text{ TeV}} dE E^{-\Gamma} \exp(-E/E_{\text{cut}}) \exp[-\tau(E, z)]. \end{aligned} \quad (\text{B16})$$

From $F_{\text{sens}}^{\text{CTA}}(z)$, it is therefore possible to determine the value of $A_{\text{th}}(z)$, which allows us to determine the threshold luminosity by means of Eq. (B14) as

$$\begin{aligned} L_{\text{th}}(z) &= 4\pi d_L^2(z) F_{\text{sens}}^{\text{CTA}}(z) \\ &\times \frac{\int_{0.1 \text{ GeV}/(1+z)}^{100 \text{ GeV}/(1+z)} dE E^{-\Gamma+1} \exp(-E/E_{\text{cut}})}{\int_{30 \text{ GeV}}^{30 \text{ TeV}} dE E^{-\Gamma} \exp(-E/E_{\text{cut}}) \exp[-\tau(E, z)]}. \end{aligned} \quad (\text{B17})$$

The HSP and LISP point source sensitivity has been obtained through extensive Monte Carlo simulations [1,53] and is shown in Fig. 13 for 3 hrs of observation across different energy bins. The same plot compares the differential sensitivity with the γ -ray spectra of LISP and HSP blazars normalized to the minimum detectable intensity. As shown, the spectra observed at Earth are redshift dependent, as sources at different redshifts experience different levels of EBL absorption. Figure 14 instead shows the integrated HSP and LISP point source sensitivity in the energy range 30 GeV to 30 TeV as a function of z , where the normalization of the SED in Eq. (B13) has been fixed to its threshold value $A_{\text{th}}(z)$, obtained from Eq. (B16) by employing the Monte Carlo modeling of $F_{\text{sens}}^{\text{CTA}}(z)$.

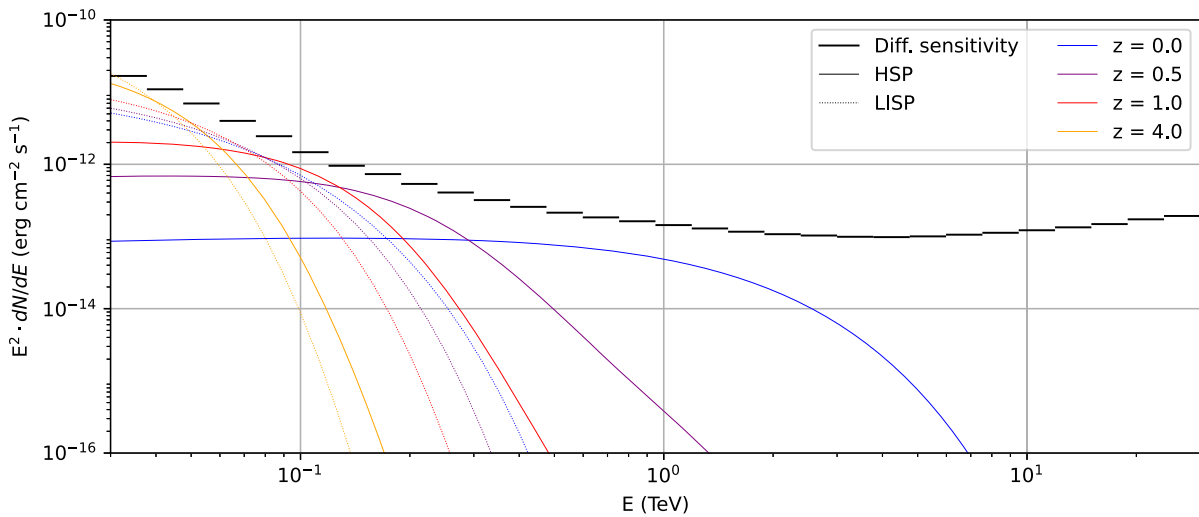


FIG. 13. Differential flux sensitivity for 5 hrs of observation in the considered CTAO energy range (thick black horizontal dashes), compared with the energy spectra of HSP and LISP sources (solid and dotted lines, respectively) at four different redshifts. The spectra are normalized to the minimal flux detectable with the planned EGAL survey.

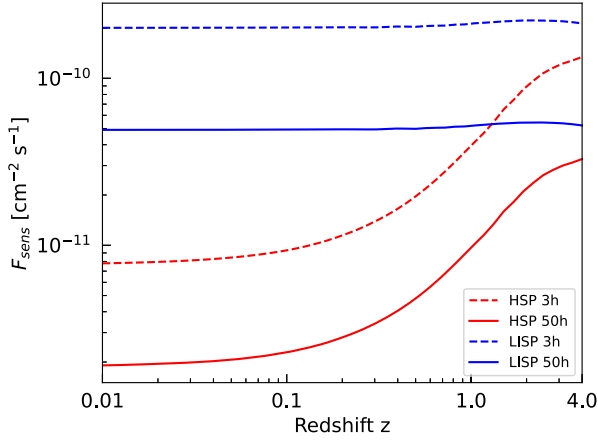


FIG. 14. CTA sensitivity $F_{\text{sens}}^{\text{CTA}}(z)$ for 3 hrs and 5 hrs of observation as a function of redshift, for the HSP and LISP γ -ray source classes.

b. Gamma-ray emission from dark matter

Window functions for dark matter are modeled as in Refs. [54–56], to which we refer for additional details. For decaying dark matter, we have

$$W(E, \chi) = \frac{dW}{dE d\chi} = \left(\frac{\Omega_{\text{DM}} \rho_c}{m_\chi} \right) \frac{\Gamma_d dN}{4\pi dE} [(1+z)/E] \exp[-\tau(E, z)], \quad (\text{B18})$$

where Ω_{DM} is the dark matter density parameter, ρ_c is the critical density of the Universe, m_χ the DM particle mass, Γ_d the DM decay rate, dN/dE the γ -ray energy spectrum per decay event, which we take from [57] for decay into $b\bar{b}$, $\tau^+\tau^-$, W^+W^- . The last term in the equation refers again to γ -ray absorption, as discussed for the astrophysical sources.

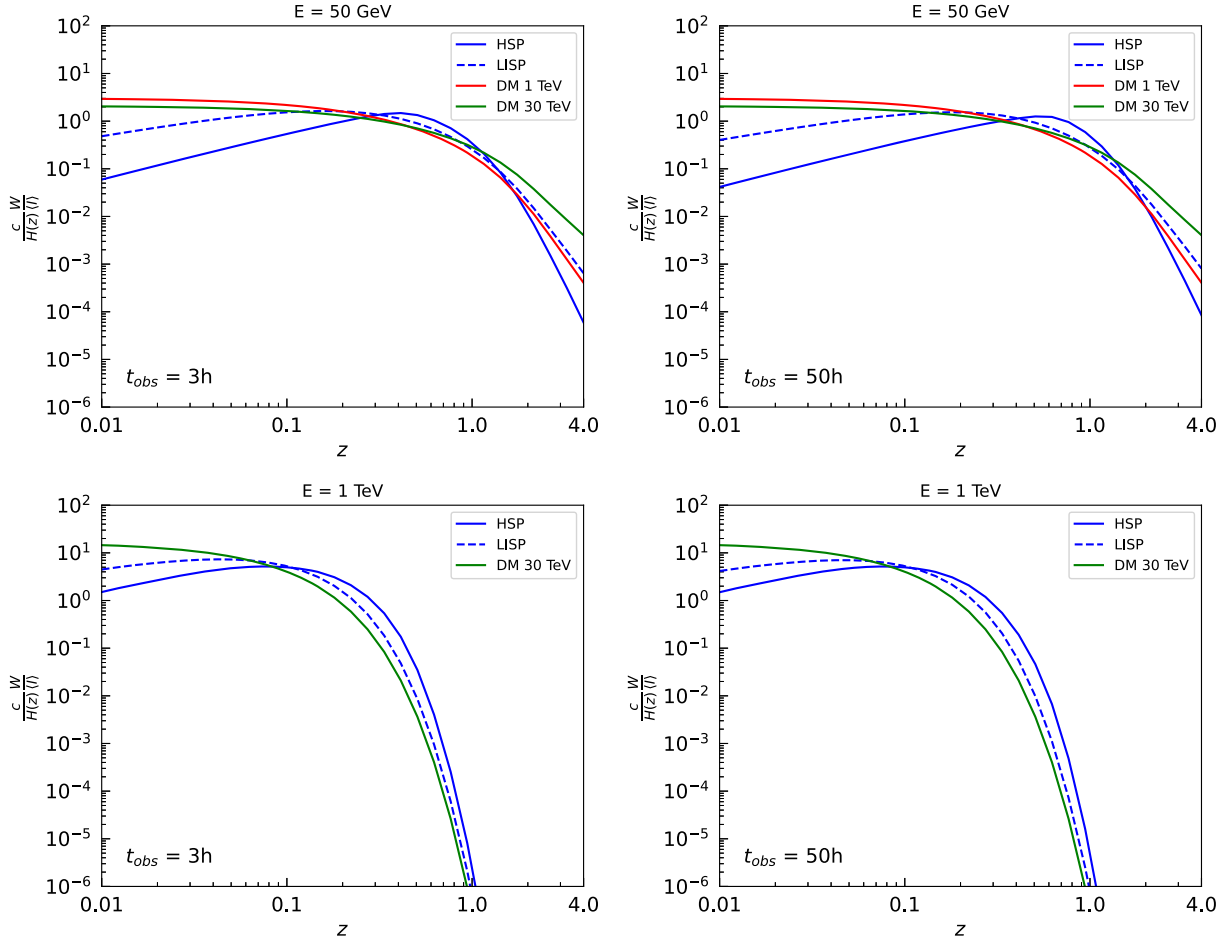


FIG. 15. γ -ray window functions, normalized to the mean γ -ray intensity, as a function of redshift (and divided by the Hubble parameter). Left panels show results for 3 hours of observation, and right panels for 50 hours. The blue solid and dashed lines represent the HSP and LISP astrophysical sources, respectively. The red and green solid lines refer to DM annihilation into $b\bar{b}$, $\langle\sigma v\rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$ and masses of 1 TeV (red) and 30 TeV (green). The upper row is calculated for a γ -ray energy of 50 GeV, while the lower row is for 1 TeV (only the 30 TeV DM case is shown here). The 3-hour observation time cases are the same as shown in Fig. 4.

For annihilating DM the window function reads

$$W(E, \chi) = \frac{dW}{dE d\chi} = \frac{1}{4\pi} \frac{\langle \sigma v \rangle}{2} \Delta^2(z) \left(\frac{\Omega_{\text{DM}} \rho_c}{m_\chi} \right)^2 (1+z)^3 \times \frac{dN}{dE} [(1+z)E] \exp[-\tau(E, z)], \quad (\text{B19})$$

where $\langle \sigma v \rangle$ is the velocity-averaged annihilation cross section. Unlike DM decay, the annihilation window function also contains the flux multiplier factor $\Delta^2(z)$ which accounts for the enhancement of the flux due to the clumping of DM and the presence of substructures within the DM halos. Details on the way we model the flux multiplier can be found in Refs. [58–60]. Window functions for both DM and astrophysical sources are plotted in Fig. 15 as function of redshift for two different energies, and for 3 hrs and 5 hrs of observation time. It can be seen that the DM window always peaks at low z , while the astrophysical signal peaks at $z \sim 0.4$ – 0.5 at 50 GeV and $z \sim 0.1$ – 0.2 at 1 TeV, with the horizon shrinking due to the effect of EBL γ -ray absorption. Figure 16, instead, displays the intensity $\langle I \rangle$ as a function of the energy for the two classes of astrophysical sources used in the analysis and for a few representative cases of DM annihilation.

c. Galaxy catalogs

The galaxy-catalog window functions are given by the redshift distribution of the number of objects in the catalog, namely,

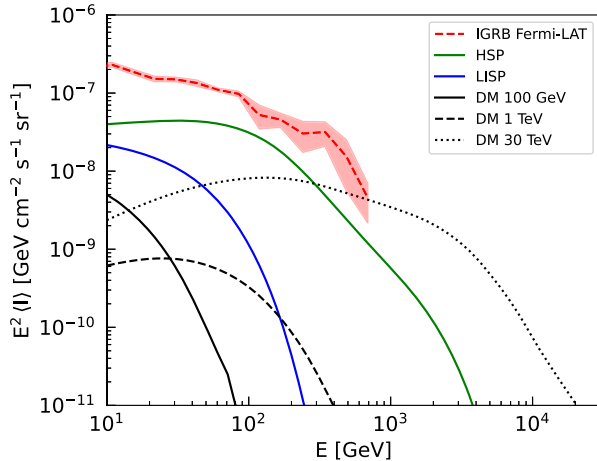


FIG. 16. γ -ray intensity in our benchmark model for HSP (solid green) and LISP (solid blue), along with three cases of dark matter annihilation for particle masses of 100 GeV (solid black), 1 TeV (dashed black), and 30 TeV (dotted black). The 100 GeV and 1 TeV lines correspond to an annihilation cross section $\langle \sigma v \rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$, while for 30 TeV we adopted $\langle \sigma v \rangle = 3 \times 10^{-23} \text{ cm}^3 \text{ s}^{-1}$. The figure also shows the measured IGRB (red dashed line and red band for its uncertainty) [61].

$$W_g(z) = \frac{dN_g}{dz}. \quad (\text{B20})$$

The window function is normalized such that $\int dz W_g(z) = 1$. Note again that $W_g(\chi)$ entering the cross-correlation APS is related to the redshift-dependent window function as $W_g(\chi) = W_g(z) \frac{dz}{d\chi} = W_g(z) H(z)/c$, where $H(z)$ is the Hubble parameter and c the speed of light. In this way we also have $\int d\chi W_g(\chi) = 1$. We use the dN_g/dz as given in [42,43] for 2MRS and [40,41] for 2 MASS.

3. Galaxies halo occupation distribution

To model the clustering of galaxies within the catalog we use the halo occupation distribution (HOD) formalism (see [62–64]). The number of galaxies residing within a DM halo of mass M is parametrized as

$$\langle N_{\text{cen}}(M) \rangle = \frac{1}{2} \left[1 + \text{erf} \left(\frac{\log M - \log M_{\text{th}}}{\sigma_{\log M}} \right) \right], \quad (\text{B21})$$

$$\langle N_{\text{sat}}(M) \rangle = \left(\frac{M}{M_1} \right)^\alpha \exp \left(-\frac{M_{\text{cut}}}{M} \right), \quad (\text{B22})$$

where, for a halo of mass M , $\langle N_{\text{cen}}(M) \rangle$ is the average number of galaxies at the center of the halo and $\langle N_{\text{sat}}(M) \rangle$ represents the average number of satellite galaxies. The parameters $M_{\text{th}}, \sigma_{\log M}, \alpha, M_1, M_{\text{cut}}$ characterize the HOD and we chose them according to [43] for 2MRS and [56] for 2 MASS.

4. 3D power spectra

The 3D power spectra $P(k)$ (PS) have been calculated by using the halo model formalism [64]. We have derived the autocorrelation 3D PS of decaying DM $P_\delta(k)$, annihilating DM $P_{\delta^2}(k)$ and astrophysical sources $P_\gamma(k)$, and the 3D cross-correlation PS among the same three categories of tracers: $P_{g\delta}(k)$ for the cross-correlation between galaxies and decaying DM γ -ray emission, $P_{g\delta^2}(k)$ for the cross-correlation between galaxies and annihilating DM γ -ray emission, for the cross-correlation between galaxies and astrophysical sources γ -ray emission $P_{g\gamma}(k)$. The explicit formulas for the power spectra are reported in Ref. [56], to which we refer the reader for details.

Figures 17 and 18 report the plots of the various 3D power spectra adopted in the analysis. Figures 19 and 20 show a few samples of auto and cross-correlation APS as a function of the multipole ℓ . It can be seen that the auto APS of astrophysical sources is constant in ℓ , since they are point-like, while the DM APS presents some structure due to the extension of the DM halos. Figure 21, instead, shows the APS as a function of energy at a fixed multipole $\ell = 100$. For comparison, the auto and cross-correlation

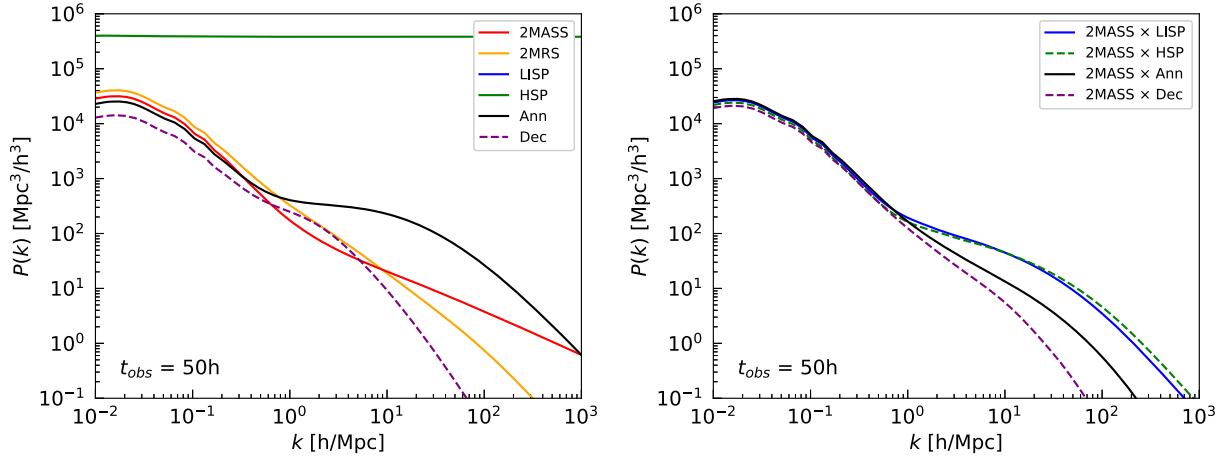


FIG. 17. Left: Autocorrelation 3D power spectra for 2 MASS and 2 MRS galaxies, LISP and HSP astrophysical sources, and annihilating and decaying dark matter. Right: Cross-correlation 3D power spectra for the cross-correlation between 2 MASS galaxies and the various γ -ray source types considered in our analysis: LISP, HSP, annihilating and decaying dark matter.

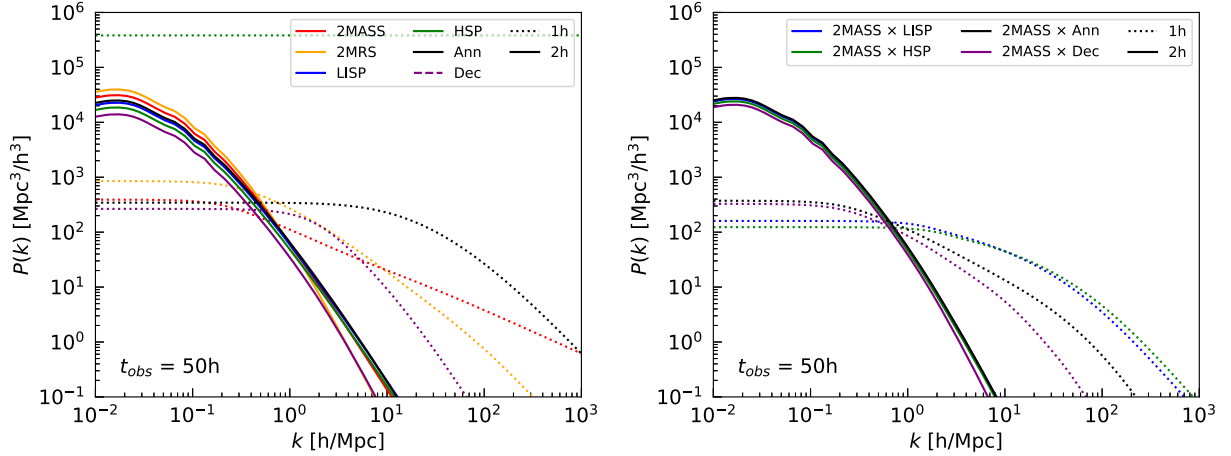


FIG. 18. The same as Fig. 17 but with the 1-halo (dotted) and the 2-halo (solid) terms shown separately.

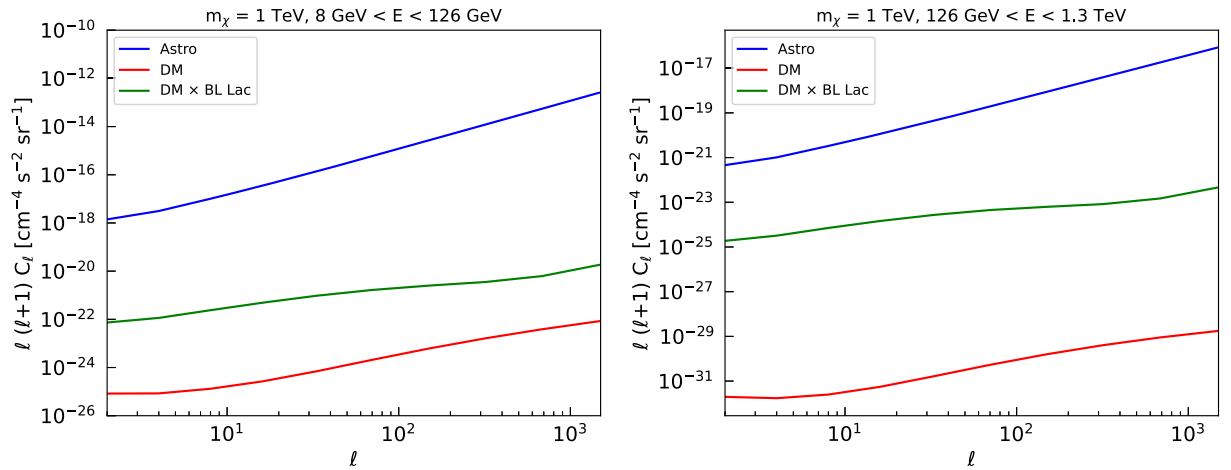


FIG. 19. Angular power spectra of the γ -ray autocorrelation as a function of the multipole, for two different energy bins. The upper (blue) curve refers to the autocorrelation among astrophysical sources, the median (green) and lower (red) curves to the correlation between sources and dark matter and the autocorrelation of dark matter, respectively, for a dark matter mass of 1 TeV and annihilation into $b\bar{b}$ with a cross section $\langle\sigma v\rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$.

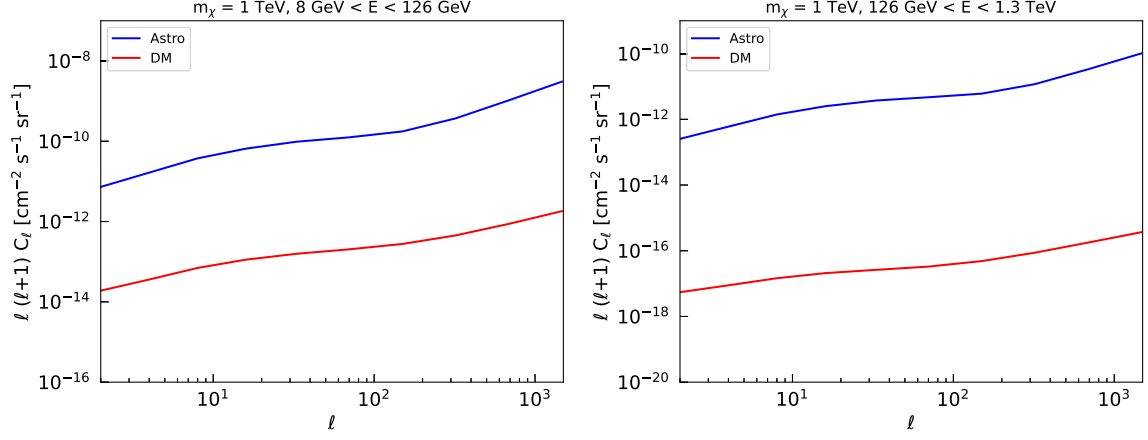


FIG. 20. Angular power spectra of the γ -ray cross-correlation with 2 MASS galaxies, as a function of the multipole, for two different energy bins. The upper (blue) curve refers to the cross-correlation with astrophysical sources, while the lower (red) curves represent the correlation with a dark matter signal produced by a particle with mass of 1 TeV and annihilation cross section into $b\bar{b}$ of $\langle\sigma v\rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$.

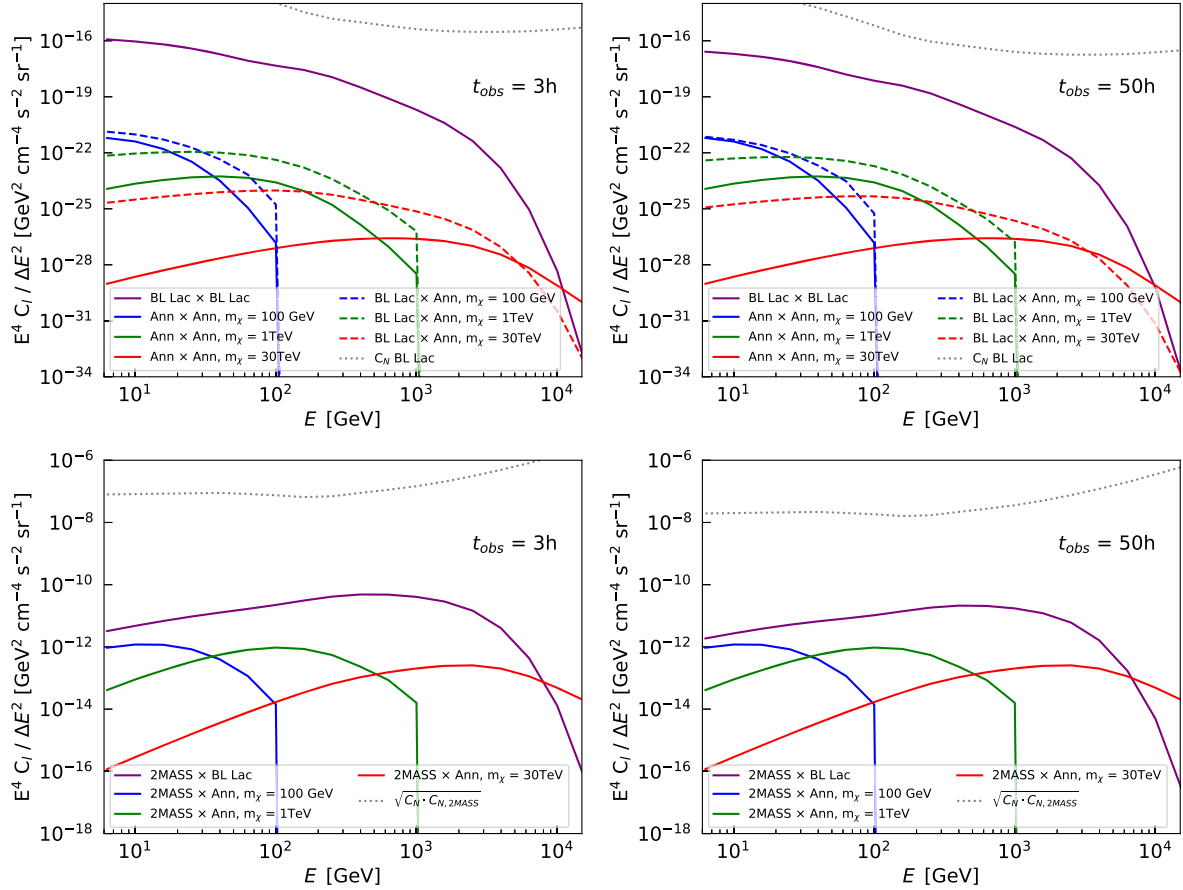


FIG. 21. Angular power spectra C_ℓ as a function of energy for $\ell = 100$. The left panels correspond to 3 hours of observation, while the right panels represent 50 hours. Dark matter signals are calculated for annihilation into $b\bar{b}$, for $\langle\sigma v\rangle = 3 \times 10^{-26} \text{ cm}^3 \text{ s}^{-1}$, and three representative masses: 100 GeV, 1 TeV and 30 TeV. The solid and dashed lines show the autocorrelation and the cross-correlation C_ℓ for various combination of γ -ray emitters and galaxy catalogs, as labelled. The black dotted lines in the topmost part of the panels represent the noise level for each category: for the upper row (γ -ray autocorrelations), the noise is dominated by the photon noise of astrophysical sources, and for the lower row (cross-correlation of γ -ray with galaxies), the geometric mean of the photon and galaxy noise is shown as a representative noise level.

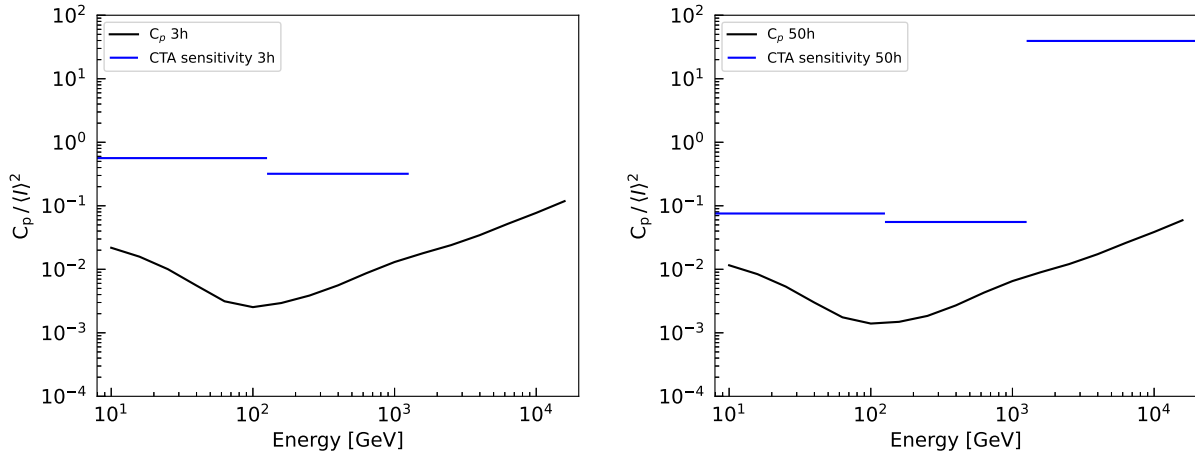


FIG. 22. CTA sensitivity to the Poisson anisotropy C_p (normalized to the square of the mean γ -ray intensity) in the low, mid and high energy bins, for 3 hours (left panel) and 50 hours (right panel) of observation. The predicted sensitivity is to the expected C_p from *unresolved* blazars (solid black lines). The reduced sensitivity at high energies is driven by the limited number of γ -ray events in this energy range.

noise level is plotted, thus showing that the expected signal-to-noise ratio is typically low.

5. Poisson anisotropy

The autocorrelation angular power spectrum of blazars is dominated by the constant 1-halo term, and is therefore flat in ℓ . It is thus convenient to define the Poisson anisotropy as the normalization of this spectrum. Figure 22 shows the Poisson anisotropy of unresolved blazars, divided by the square of the intensity as function of energy for 3 hours and 50 hours of observation. As expected, the anisotropy decreases with increasing observation time, since a larger fraction of the brighter sources (which contribute more significantly to the anisotropy) become resolved. Across the entire energy range, the anisotropy remains at the level of percent or below.

The same figure also shows the CTA sensitivity to the Poisson anisotropy in three energy bins. To derive this sensitivity, we adopt the same methodology used for setting the DM limits. In this case, however, the null hypothesis corresponds to pure noise, that is the absence of any source contribution, while the alternative hypothesis assumes the presence of a source class characterized by a Poisson anisotropy. We further assume that the energy dependence of the anisotropy of this source class is proportional to $\langle I \rangle^2$, i.e., it scales with the square of the observed intensity. As shown in the figure, the resulting sensitivity is at the level of approximately 10%.

APPENDIX C: FURTHER RESULTS

In Figs. 23–28, we show additional results on the constraints for the DM annihilation cross section and the

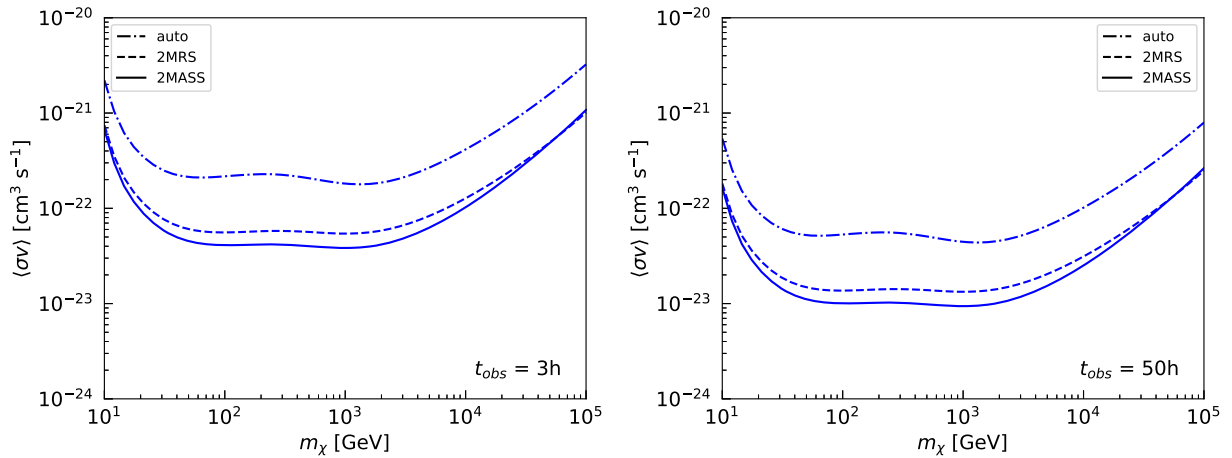


FIG. 23. Predicted upper bounds on the dark matter cross section $\langle \sigma v \rangle$, for annihilation into $b\bar{b}$, for 3 hours (left panel) and 50 hours (right panel) of CTAO observations. The bounds are derived from γ -ray autocorrelation (dot-dashed), and cross-correlation with 2 MRS (dashed lines) and 2 MASS (solid) galaxy catalogs.

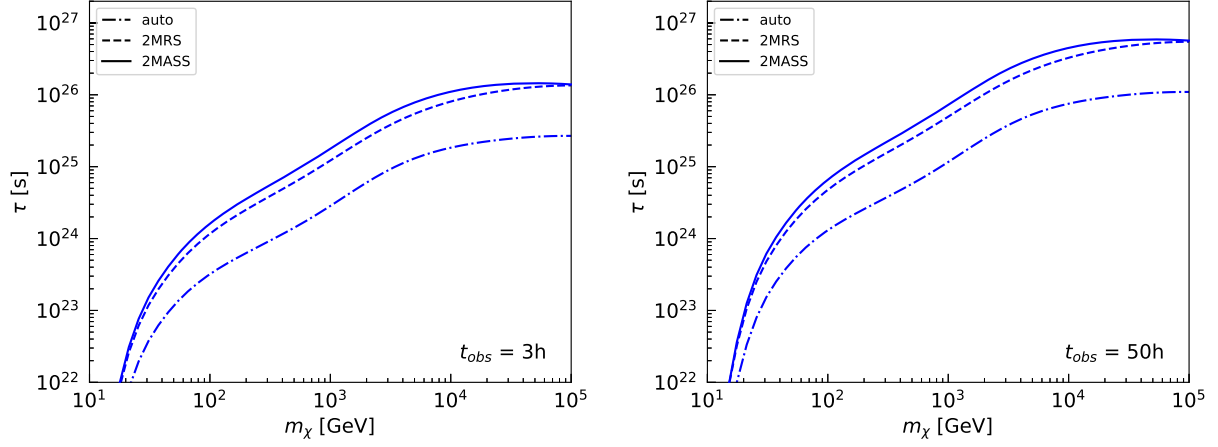


FIG. 24. Predicted lower bounds on the dark matter lifetime τ , for decay into $b\bar{b}$, for 3 hours (left panel) and 50 hours (right panel) of CTAO observations. The bounds are derived from γ -ray autocorrelation (dot-dashed), and cross-correlation with 2 MRS (dashed lines) and 2 MASS (solid) galaxy catalogs.

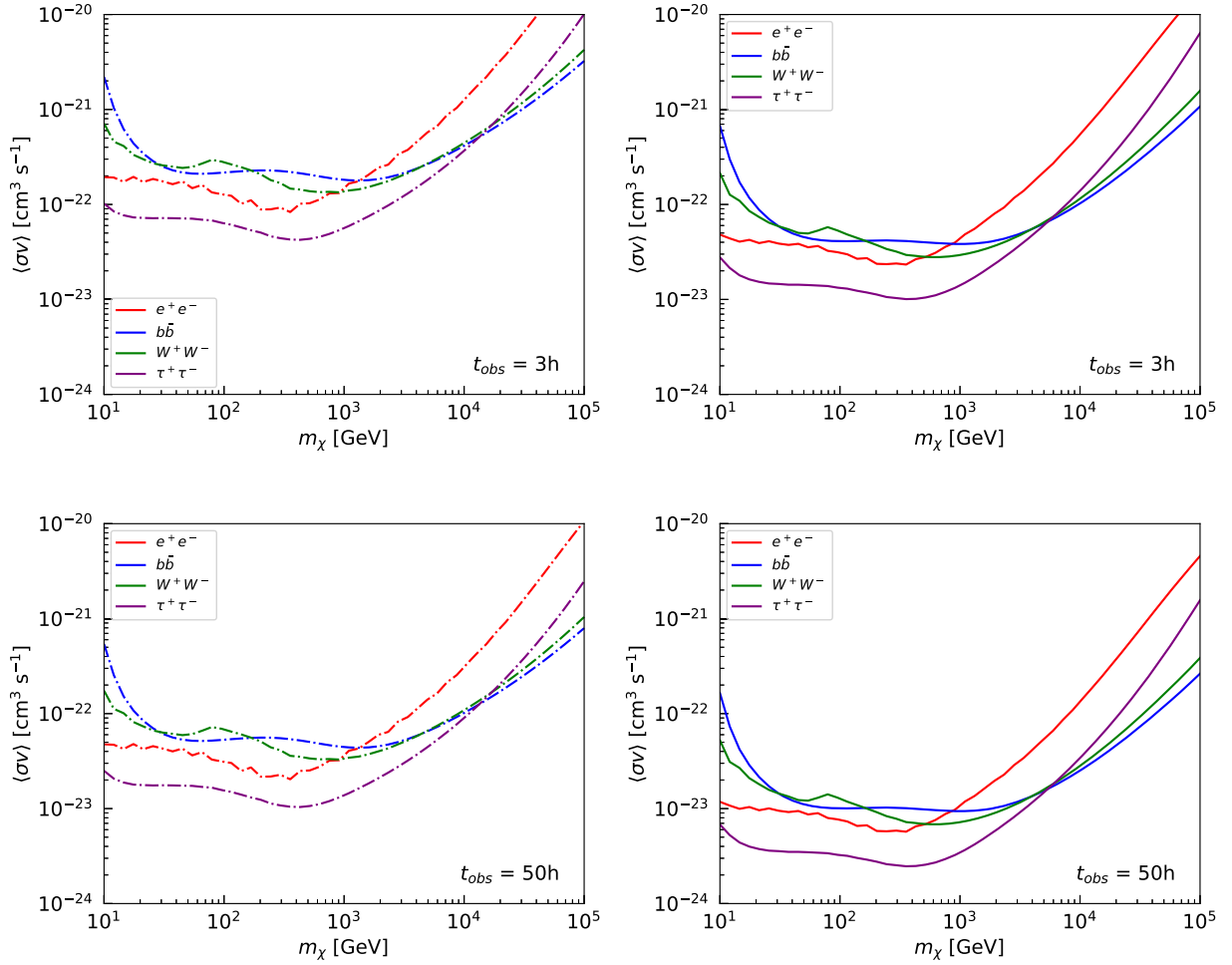


FIG. 25. Predicted upper bounds on the dark matter cross section $\langle\sigma v\rangle$, for annihilation into various final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. The upper row shows results for 3 hours of observation; the lower row for 50 hours. Left panels: bounds from γ -ray autocorrelation. Right panels: Bounds from cross-correlation with 2 MASS galaxies (the bottom-right panel reproduces the left panel of Fig. 6).

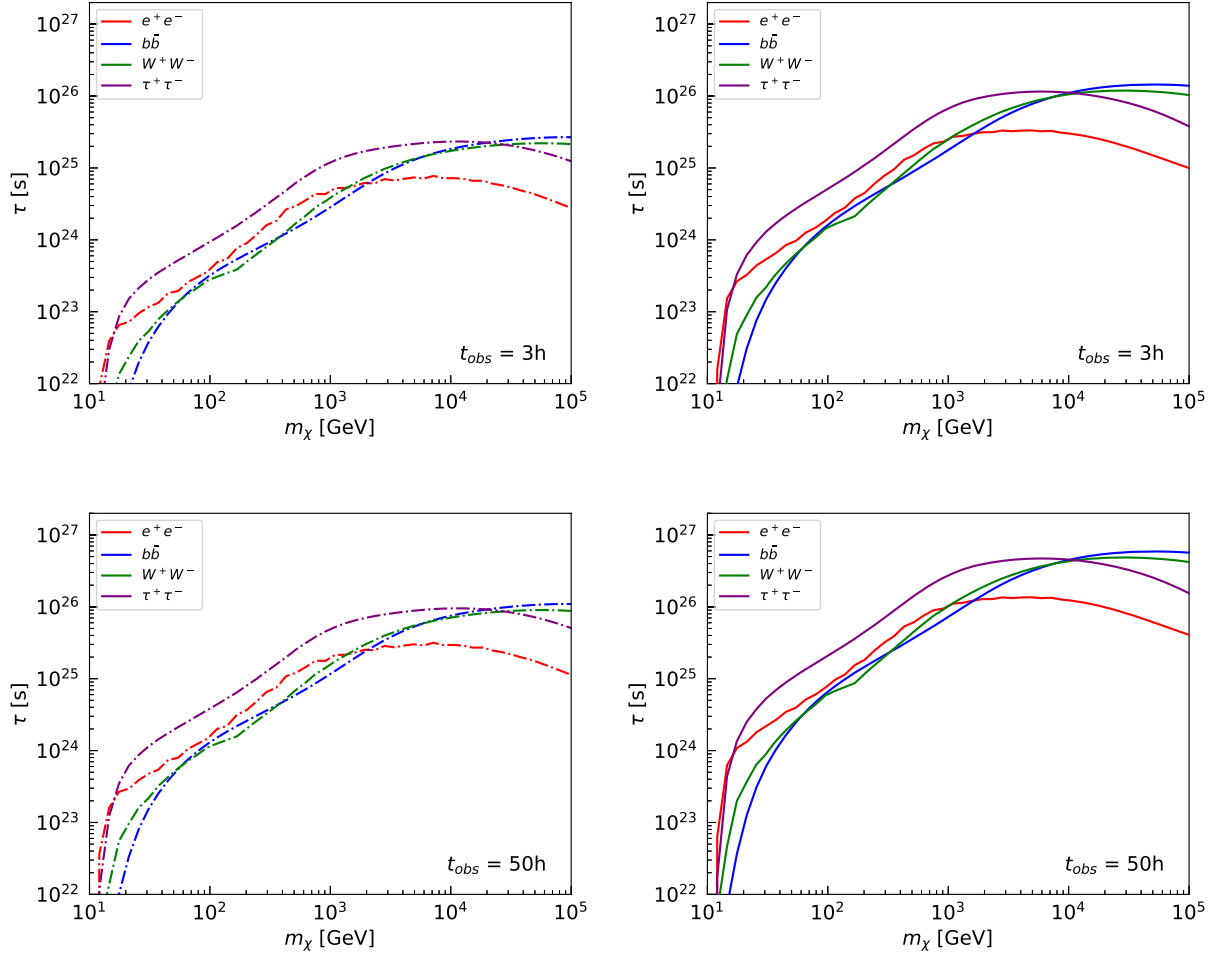


FIG. 26. Predicted lower bounds on the dark matter lifetime τ , for decay into various final states: e^+e^- , $b\bar{b}$, W^+W^- and $\tau^+\tau^-$. The upper row shows results for 3 hours of observation; the lower row for 50 hours. Left panels: Bounds from the γ -ray autocorrelation. Right panels: bounds from cross-correlation with 2 MASS galaxies (the bottom-right panel reproduces the right panel of Fig. 6).

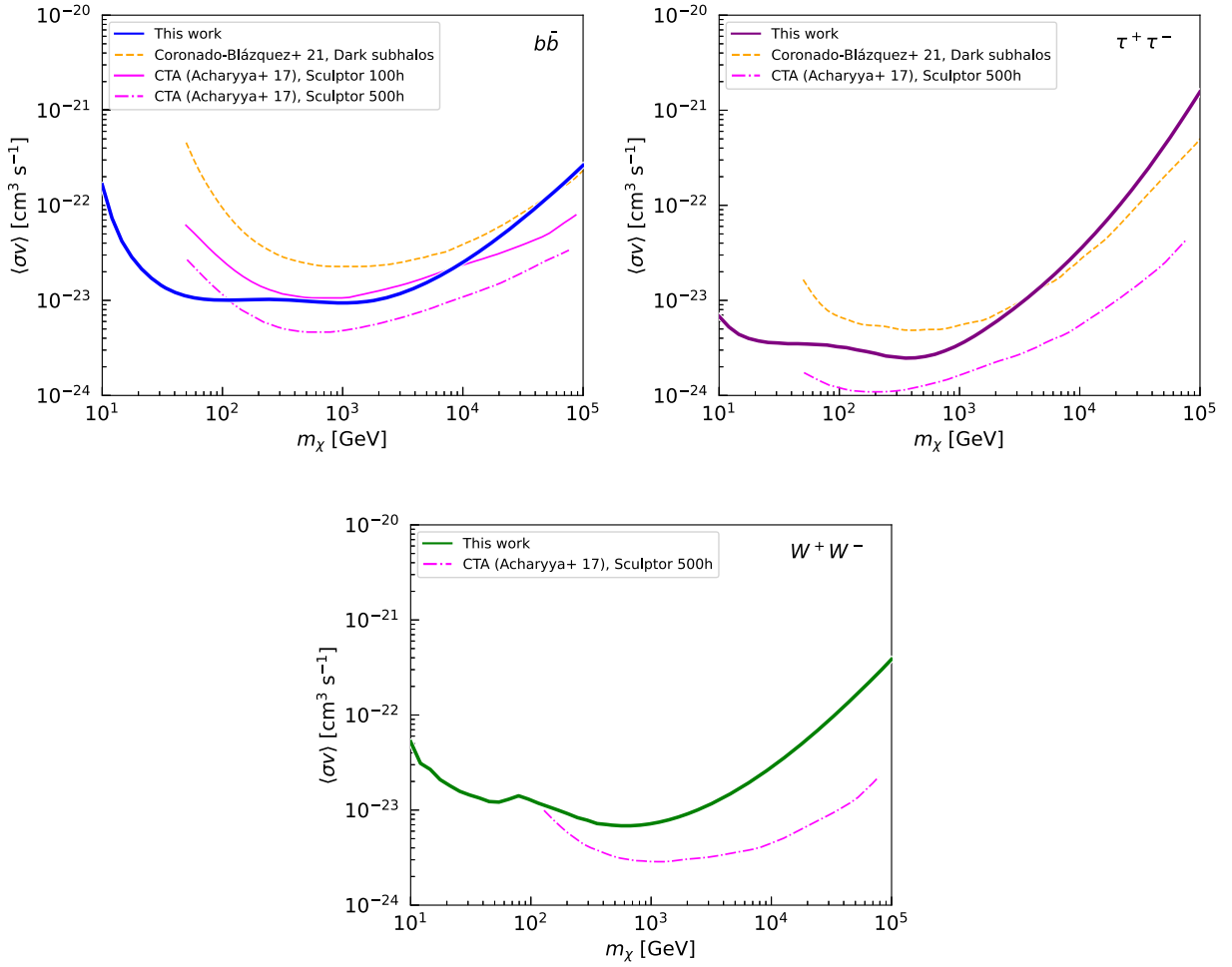


FIG. 27. Comparison of the predicted upper bounds on the dark matter annihilation cross section $\langle\sigma v\rangle$, with existing constraints from the literature: bounds from nonobservation of dark subhalos (dashed yellow) [35], from the Sculptor dwarf galaxy [44] with 100 hours (solid magenta) and 500 hours (dot-dashed magenta) of observations with CTAO. Three channels are displayed: $b\bar{b}$ (upper left panel), $\tau^+\tau^-$ (upper right) and the W^+W^- (lower). The case for the $b\bar{b}$ channel is also shown in Fig. 5.

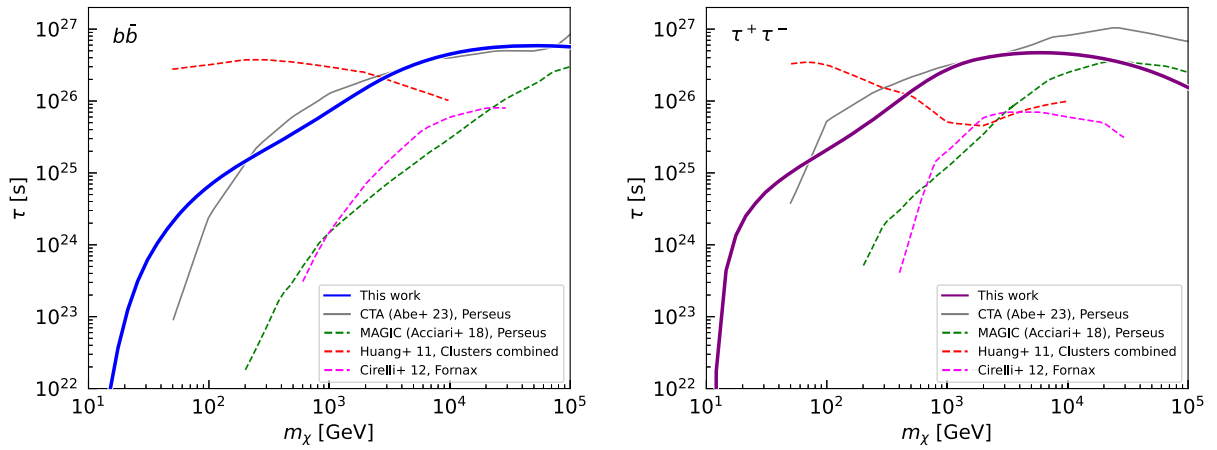


FIG. 28. Comparison of the predicted lower bounds on the DM particle lifetime τ for the $b\bar{b}$ and $\tau^+\tau^-$ decay channels, with existing constraints from the literature: limits from Perseus with CTAO [45] and MAGIC [46], from galaxy clusters [47], and from Fornax [48]. The case for the $b\bar{b}$ channel is also shown in Fig. 5.

decay lifetime, exploring various annihilation and decay channels, as well as different observation times. We also provide further comparisons with existing bounds from the literature.

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